upGrad

Module : Inferential Statistics

Instructor: Dr. Darshan Ingle







What you'll learn

1. Probability
2. Random Variable
3. Binomial Distribution
4. Normal Distribution
5. PDF vs CDF
6. Standard Score
7. CLT and Baye's Theorem

Xpert: Dr. Darshan Ingle

Random Variables

- Example: Two dice are tossed.
- The Random Variable is **X** = "The sum of the scores on the two dice".



Let's make a table of all possible values:

		1st Die					
		1	**2**	**3**	**4**	**5**	**6**
2nd Die	**1**	2	3	4	5	6	7
	2	3	4	5	6	7	8
	3	4	5	6	7	8	9
	4	5	6	7	8	9	10
	5	6	7	8	9	10	11
	6	7	8	9	10	11	12

There are **6 × 6 = 36** possible outcomes, and the Sample Space (which is the sum of the scores on the two dice) is {2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12}

Let's count how often each value occurs, and work out the probabilities:

- 2 occurs just once, so P(X = 2) = 1/36
- 3 occurs twice, so P(X = 3) = 2/36 = 1/18
- 4 occurs three times, so P(X = 4) = 3/36 = 1/12
- 5 occurs four times, so P(X = 5) = 4/36 = 1/9
- 6 occurs five times, so P(X = 6) = 5/36
- 7 occurs six times, so P(X = 7) = 6/36 = 1/6
- 8 occurs five times, so P(X = 8) = 5/36
- 9 occurs four times, so P(X = 9) = 4/36 = 1/9
- 10 occurs three times, so P(X = 10) = 3/36 = 1/12
- 11 occurs twice, so P(X = 11) = 2/36 = 1/18
- 12 occurs just once, so P(X = 12) = 1/36

Probability Distribution

- A probability distribution is a table or an equation that links each outcome of a statistical experiment with its probability of occurrence.
- Example to study relationship between random variables and probability distributions.
- Suppose you flip a coin two times.
- We get four possible outcomes: HH, HT, TH, and TT.
- Now, let the variable X represent the number of Heads that result from this experiment.
- The variable X can take on the values 0, 1, or 2.
- In this example, X is a random variable; because its value is determined by the outcome of a statistical experiment.

H/T → H/T

The table represents the probability distribution of the random variable X.

Number of heads	Probability
0	0.25
1	0.50
2	0.25

Cumulative Probability Distributions

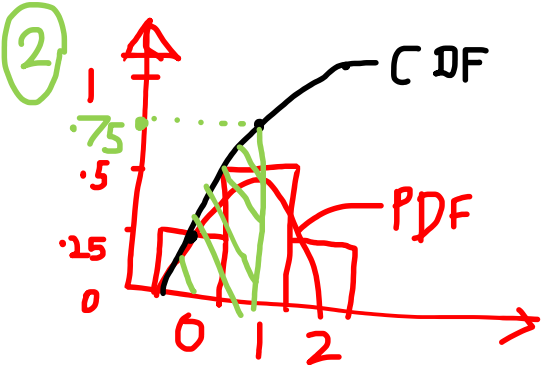
- A **cumulative probability** refers to the probability that the value of a random variable falls within a specified range.
- Example: Coin flip experiment. If we flip a coin two times, we might ask:
- What is the probability that the coin flips would result in one or fewer heads? $P(X \leq 1)$
- The answer would be a cumulative probability.

1

$$P(X \leq 1) = P(X = 0) + P(X = 1) = 0.25 + 0.50 = 0.75$$

- Like a probability distribution, a cumulative probability distribution can be represented by a table or an equation.

2



Number of heads: x	Probability: P(X = x)	Cumulative Probability: P(X ≤ x)
0	0.25	0.25
1	0.50	0.75
2	0.25	1.00

Uniform Probability Distribution

- The simplest probability distribution occurs when all of the values of a random variable occur with equal probability. This probability distribution is called the **uniform distribution**.

Uniform Distribution. Suppose the random variable X can assume k different values. Suppose also that the $P(X = x_k)$ is constant. Then,

$$P(X = x_k) = 1/k$$

Example 1

Suppose a die is tossed. What is the probability that the die will land on 5?

$$\frac{1}{6}$$

Uniform Probability Distribution

Example 2

Suppose we repeat the dice tossing experiment described in Example 1. This time, we ask what is the probability that the die will land on a number that is smaller than 5?

$$P(X<5) = P(X=1) + \dots + P(X=4)$$

$$= \frac{1}{6} + \dots + \frac{1}{6}$$

$$= \frac{4}{6} = \frac{2}{3}$$

Expected Value (EV)

- The expected value (EV) is an anticipated value for an investment at some point in the future.
- In statistics and probability analysis, the expected value is calculated by multiplying each of the possible outcomes by the likelihood each outcome will occur and then summing all of those values.
- By calculating expected values, investors can choose the scenario most likely to give the desired outcome.

$$EV = \sum P(X_i) \times X_i$$

- **Example:** Consider a normal six-sided die. Once you roll the die, it has an equal one-sixth chance of landing on one, two, three, four, five, or six. Given this information, the calculation is straightforward:

$$\left(\frac{1}{6} \times 1\right) + \left(\frac{1}{6} \times 2\right) + \left(\frac{1}{6} \times 3\right) + \left(\frac{1}{6} \times 4\right) + \left(\frac{1}{6} \times 5\right) + \left(\frac{1}{6} \times 6\right) = 3.5$$

- If you were to roll a six-sided die an infinite amount of times, you see the average value equals 3.5.

Combinations Formula:

$$C(n,r) = \frac{n!}{(r!(n-r)!)}$$

For n ≥ r ≥ 0.

Combinations Formula:

$$C(n,r) = \frac{n!}{(r!(n-r)!)}$$

Combination Problem 1
Choose 2 Prizes from a Set of 6 Prizes

You have won first place in a contest and are allowed to choose 2 prizes from a table that has 6 prizes numbered 1 through 6. How many different combinations of 2 prizes could you possibly choose?

- In this example, we are taking a subset of 2 prizes (r) from a larger set of 6 prizes (n). Looking at the formula, we must calculate "6 choose 2."
- C (6,2)= 6!/(2! * (6-2)!) = 6!/(2! * 4!) = 15 Possible Prize Combinations
- The 15 potential combinations are {1,2}, {1,3}, {1,4}, {1,5}, {1,6}, {2,3}, {2,4}, {2,5}, {2,6}, {3,4}, {3,5}, {3,6}, {4,5}, {4,6}, {5,6}

Combinations Formula:

$$C(n,r) = \frac{n!}{(r!(n-r)!)}$$

Combination Problem 2
Choose 3 Students from a Class of 25

A teacher is going to choose 3 students from her class to compete in the spelling bee. She wants to figure out how many unique teams of 3 can be created from her class of 25.

In this example, we are taking a subset of 3 students (r) from a larger set of 25 students (n). Looking at the formula, we must calculate "25 choose 3."

C (25,3)= 25!/(3! * (25-3)!)= 2,300 Possible Teams

Binomial Probability Distribution

A **binomial experiment** is a statistical experiment that has the following properties:

1. The experiment consists of *n* repeated trials.
2. Each trial can result in just two possible outcomes. We call one of these outcomes a success and the other, a failure.
3. The probability of success, denoted by *P*, is the same on every trial.
4. The trials are independent; that is, the outcome on one trial does not affect the outcome on other trials.

Consider the following statistical experiment. You flip a coin 2 times and count the number of times the coin lands on heads. This is a binomial experiment because:

1. The experiment consists of repeated trials. We flip a coin 2 times.
2. Each trial can result in just two possible outcomes - heads or tails.
3. The probability of success is constant - 0.5 on every trial.
4. The trials are independent; that is, getting heads on one trial does not affect whether we get heads on other trials.

Binomial Probability Distribution

Notation

The following notation is helpful, when we talk about binomial probability.

- *x*: The number of successes that result from the binomial experiment.
- *n*: The number of trials in the binomial experiment.
- *P*: The probability of success on an individual trial.
- *Q*: The probability of failure on an individual trial. (This is equal to 1 - *P*.)
- *n!*: The factorial of n (also known as n factorial).
- b(*x*; *n*, *P*): Binomial probability - the probability that an *n*-trial binomial experiment results in exactly *x* successes, when the probability of success on an individual trial is *P*.
- ${}_nC_r$: The number of combinations of *n* things, taken *r* at a time.

Binomial Probability Distribution

Binomial Formula. Suppose a binomial experiment consists of *n* trials and results in *x* successes. If the probability of success on an individual trial is *P*, then the binomial probability is:

$$b(x; n, P) = {}_nC_x * P^x * (1 - P)^{n - x}$$

or

$$b(x; n, P) = \{ n! / [x! (n - x)!] \} * P^x * (1 - P)^{n - x}$$

Binomial Probability Distribution

Example 1

Suppose a die is tossed 5 times. What is the probability of getting exactly 2 fours?

Solution: This is a binomial experiment in which the number of trials is equal to 5, the number of successes is equal to 2, and the probability of success on a single trial is 1/6 or about 0.167. Therefore, the binomial probability is:

$$b(2; 5, 0.167) = {}_5C_2 * (0.167)^2 * (0.833)^3$$

$$b(2; 5, 0.167) = 0.161$$

Cumulative Binomial Probability

A **cumulative binomial probability** refers to the probability that the binomial random variable falls within a specified range (e.g., is greater than or equal to a stated lower limit and less than or equal to a stated upper limit).

Example 2
The probability that a student is accepted to a prestigious college is 0.3. If 5 students from the same school apply, what is the probability that at most 2 are accepted?

Solution: To solve this problem, we compute 3 individual probabilities, using the binomial formula. The sum of all these probabilities is the answer we seek. Thus,

$$b(x \leq 2; 5, 0.3) = b(x = 0; 5, 0.3) + b(x = 1; 5, 0.3) + b(x = 2; 5, 0.3)$$

$$b(x \leq 2; 5, 0.3) = 0.1681 + 0.3601 + 0.3087$$

$$b(x \leq 2; 5, 0.3) = 0.8369$$

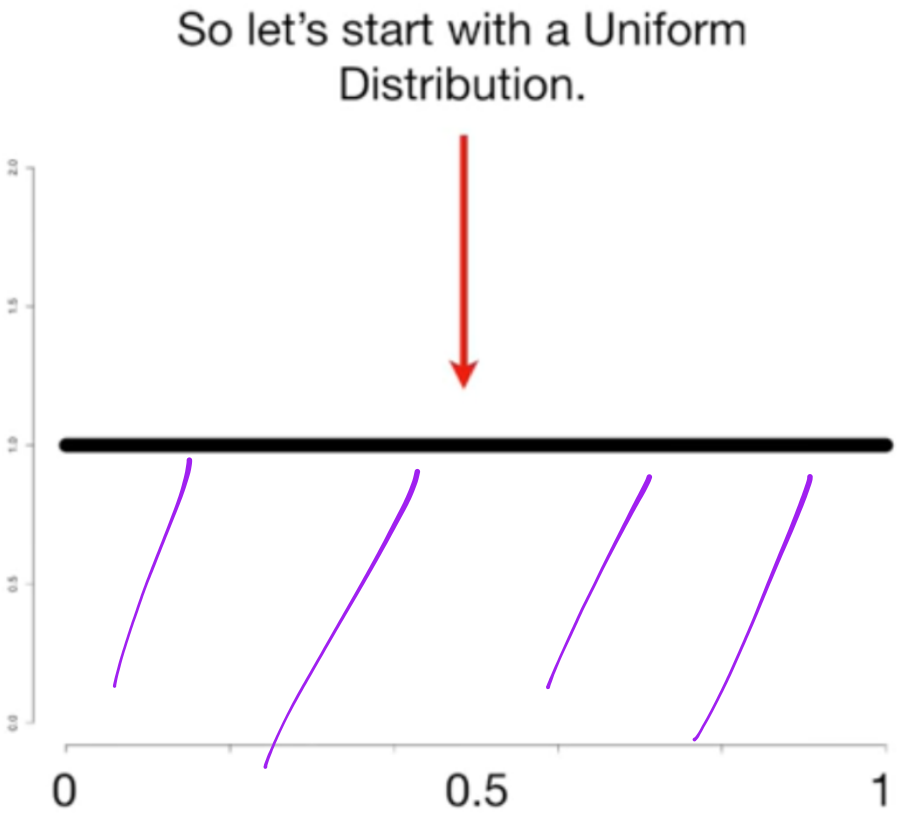
Central Limit Theorem



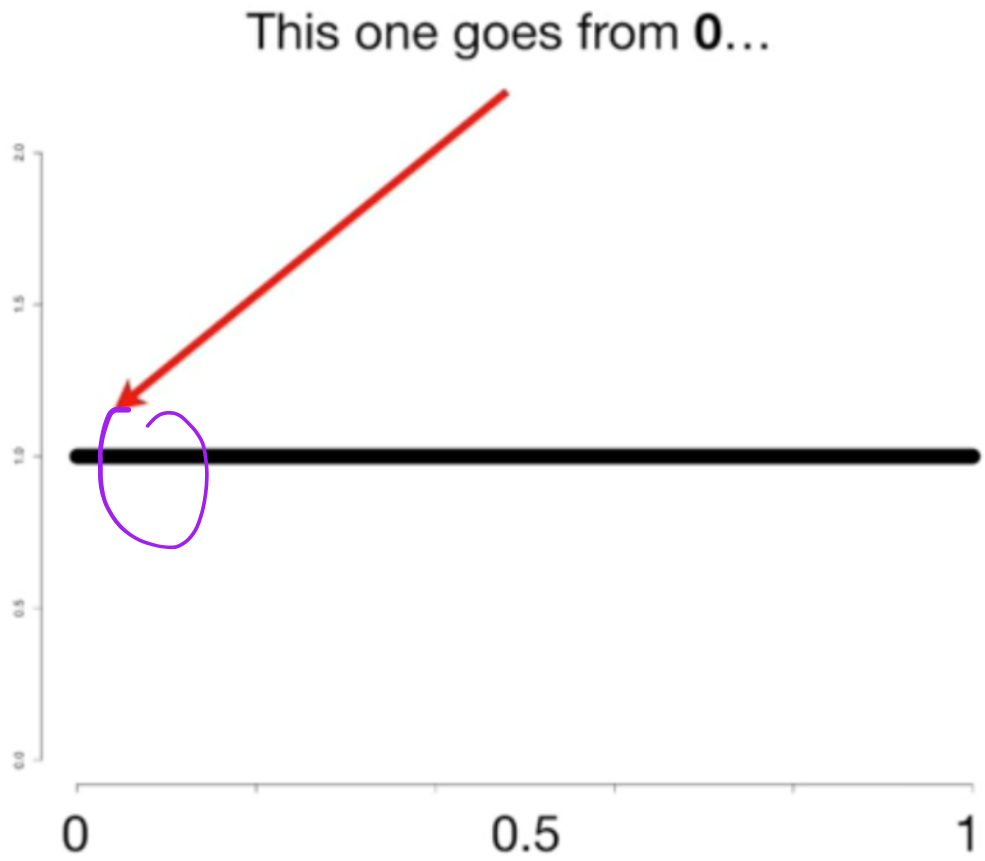
The Central Limit Theorem is the basis for a lot of statistics and the good news is that it is a pretty simple concept.

Like most things in statistics, I think The Central Limit Theorem is easiest to understand if we look at some examples.

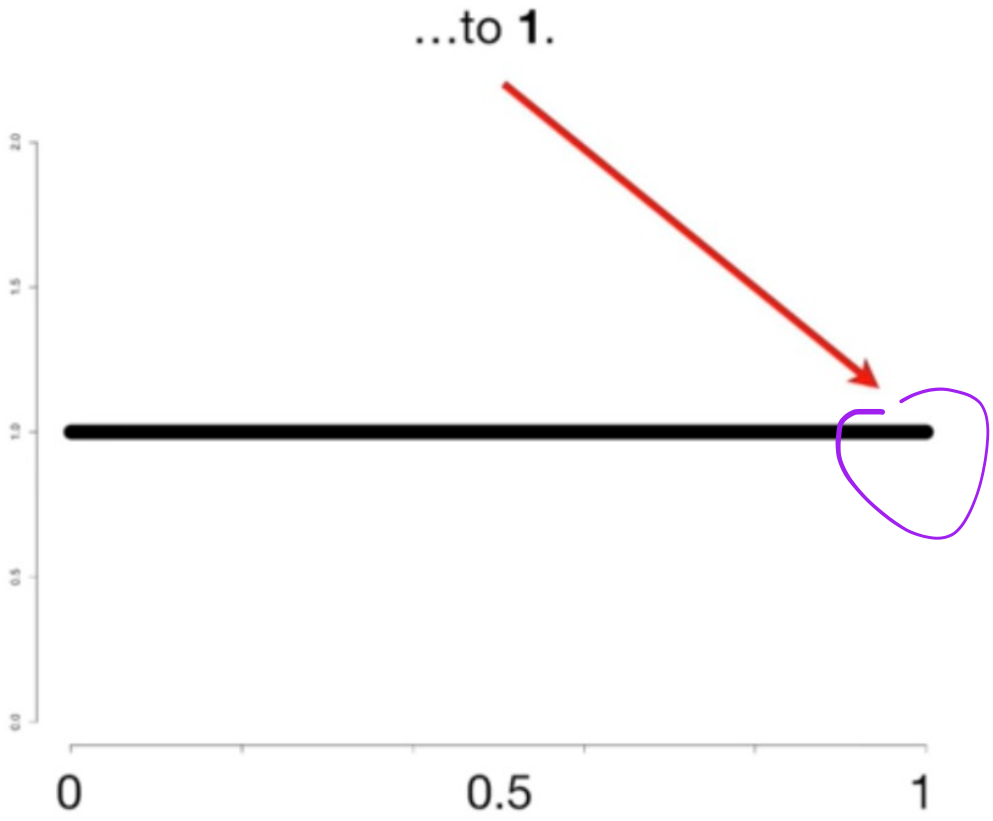
Central Limit Theorem



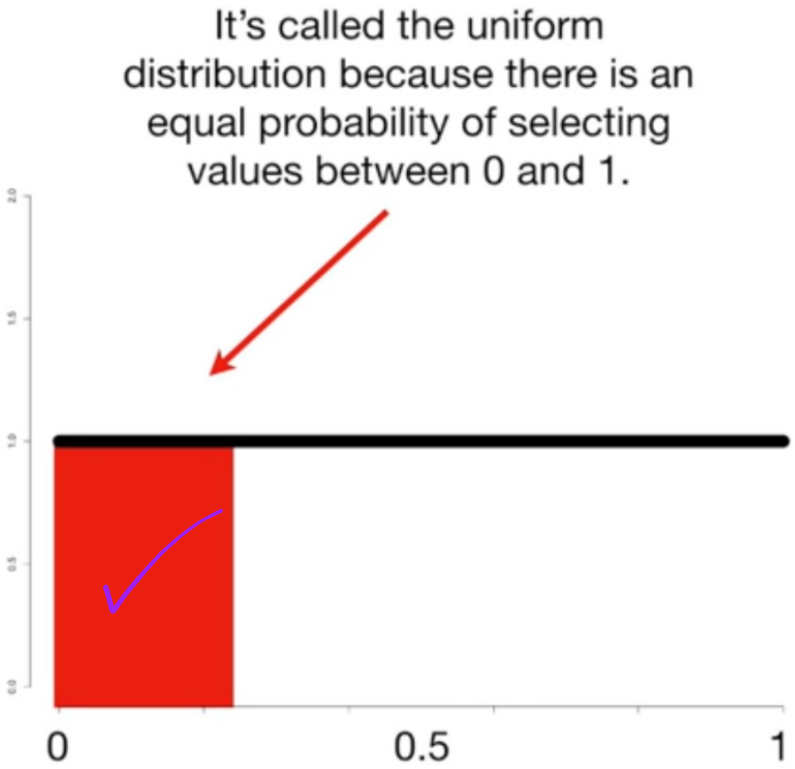
Central Limit Theorem



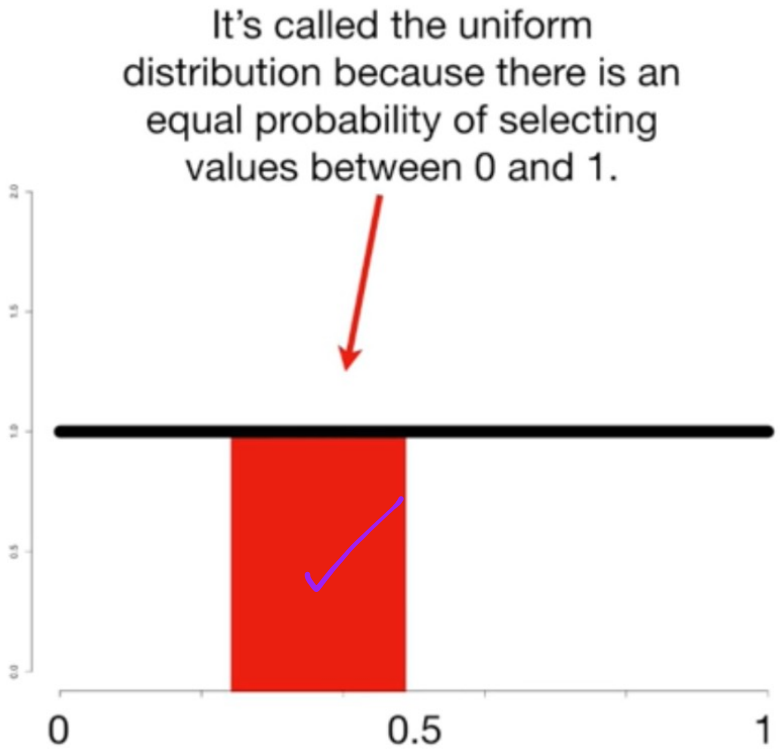
Central Limit Theorem



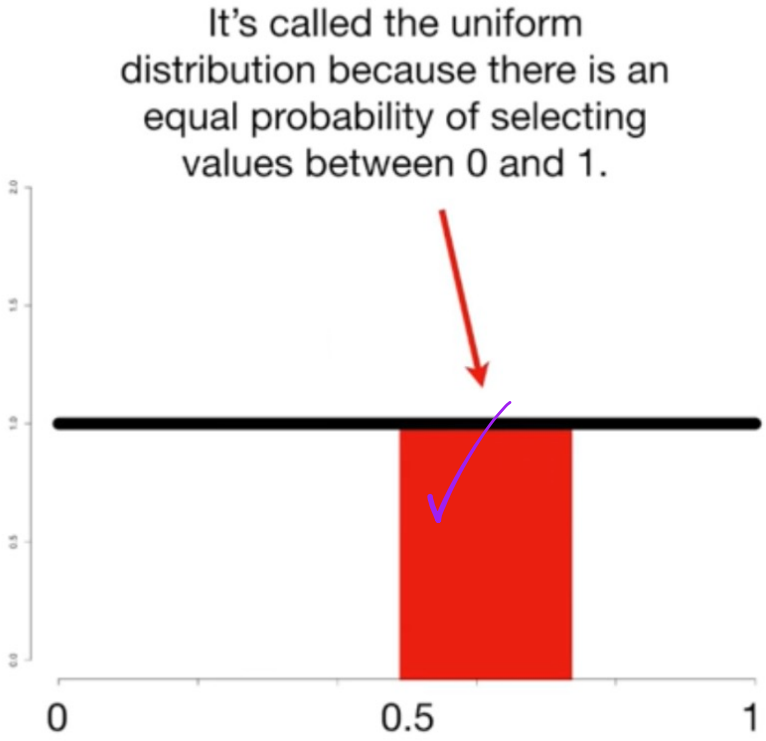
Central Limit Theorem



Central Limit Theorem

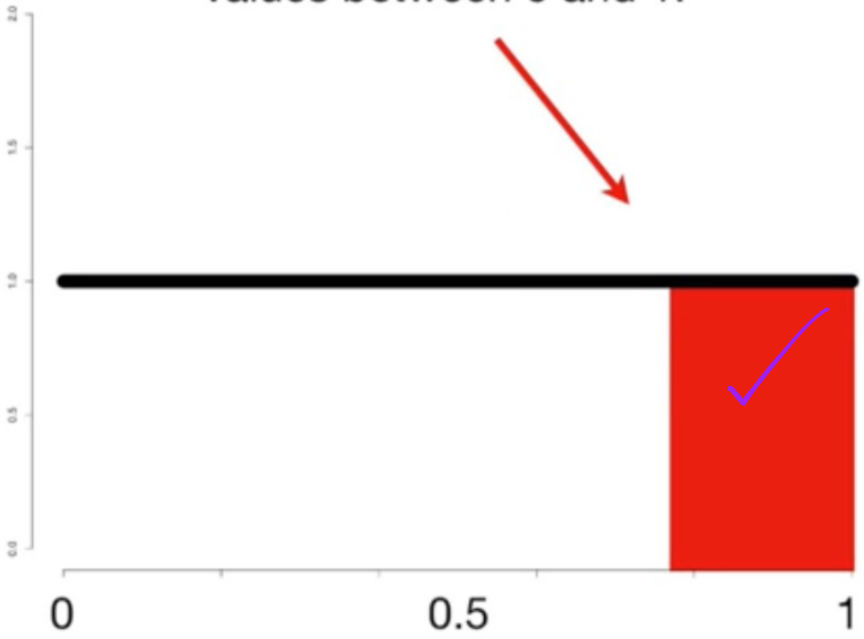


Central Limit Theorem



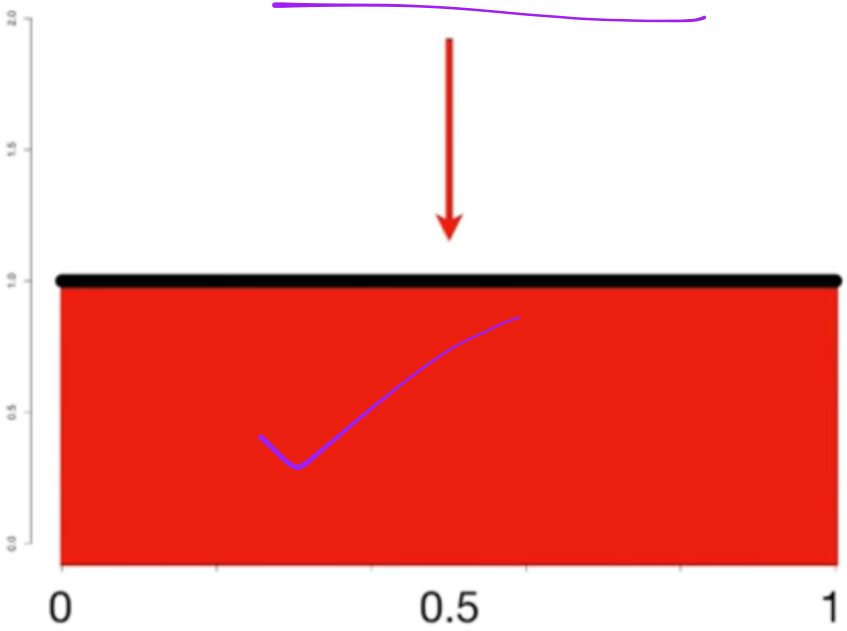
Central Limit Theorem

It's called the uniform distribution because there is an equal probability of selecting values between 0 and 1.



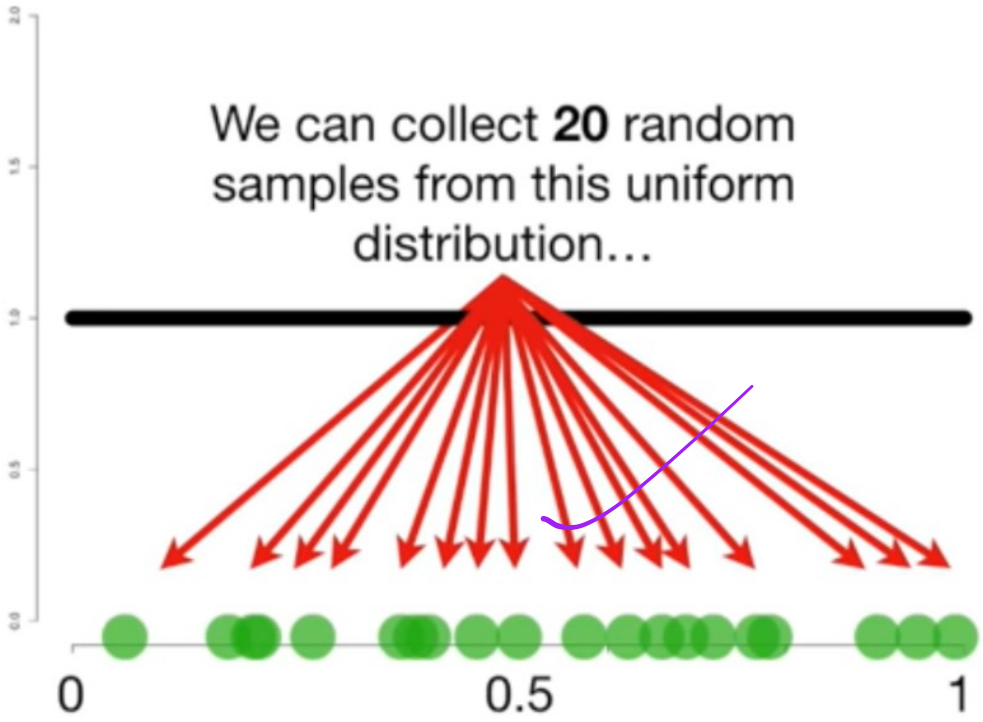
Central Limit Theorem

The probabilities are all equal,
and thus, are “uniform”.

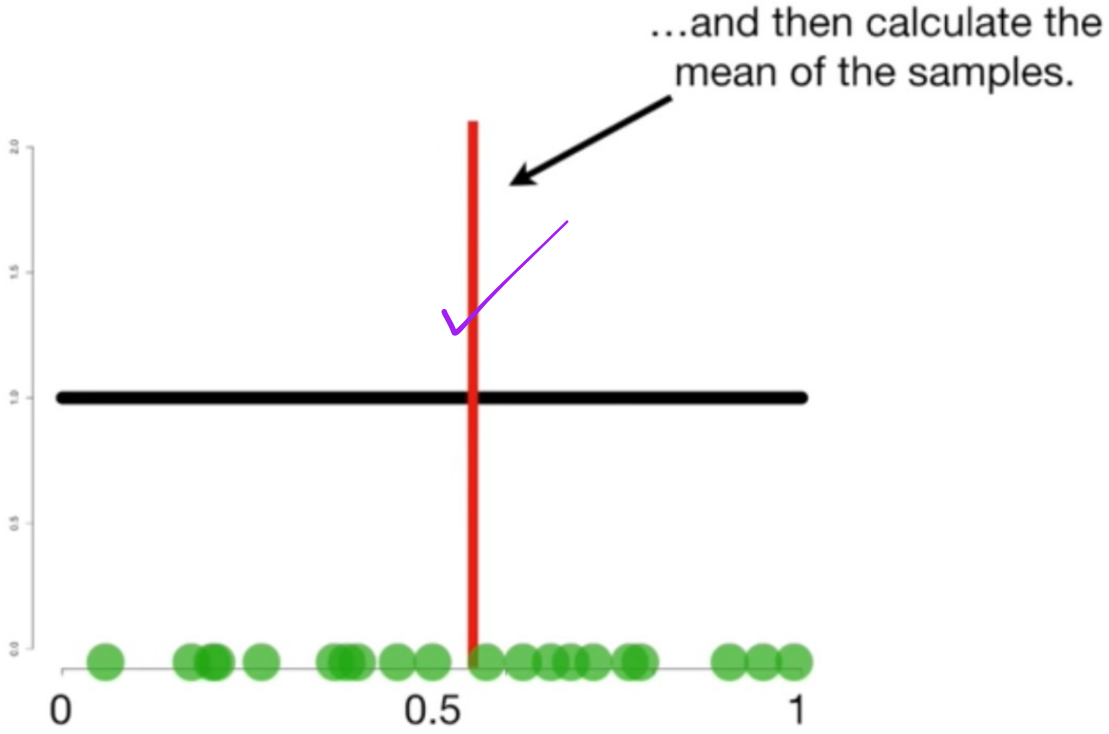


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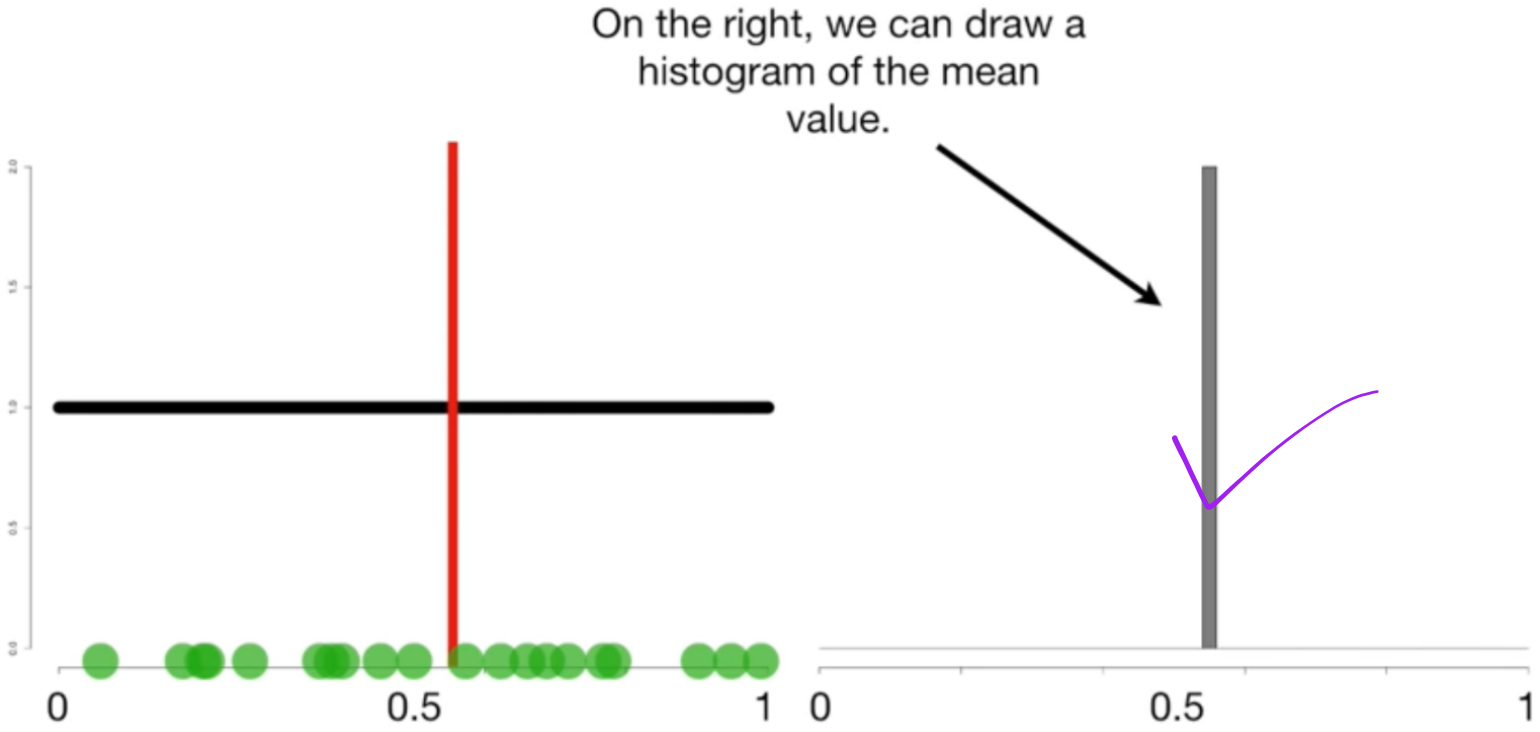
Central Limit Theorem



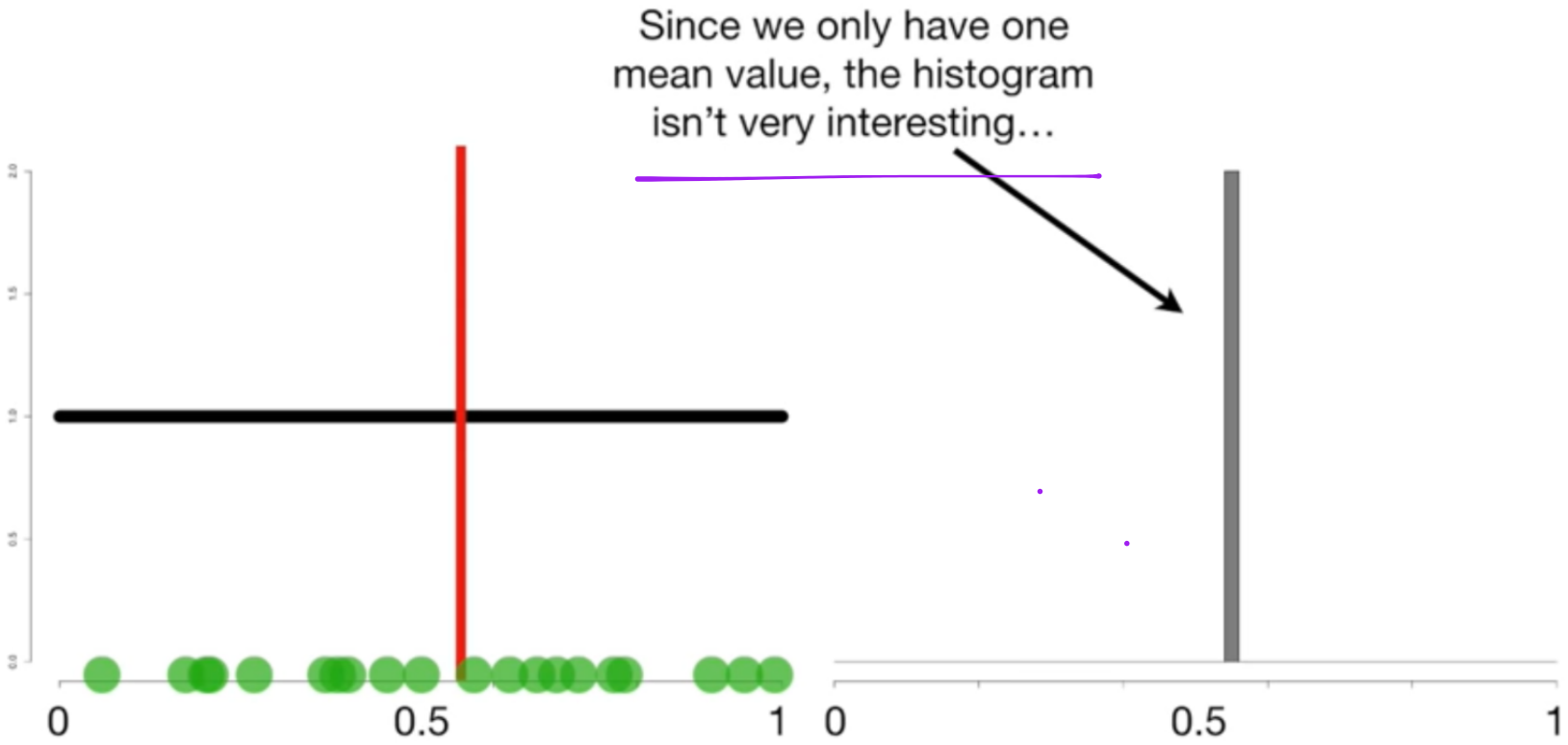
Central Limit Theorem



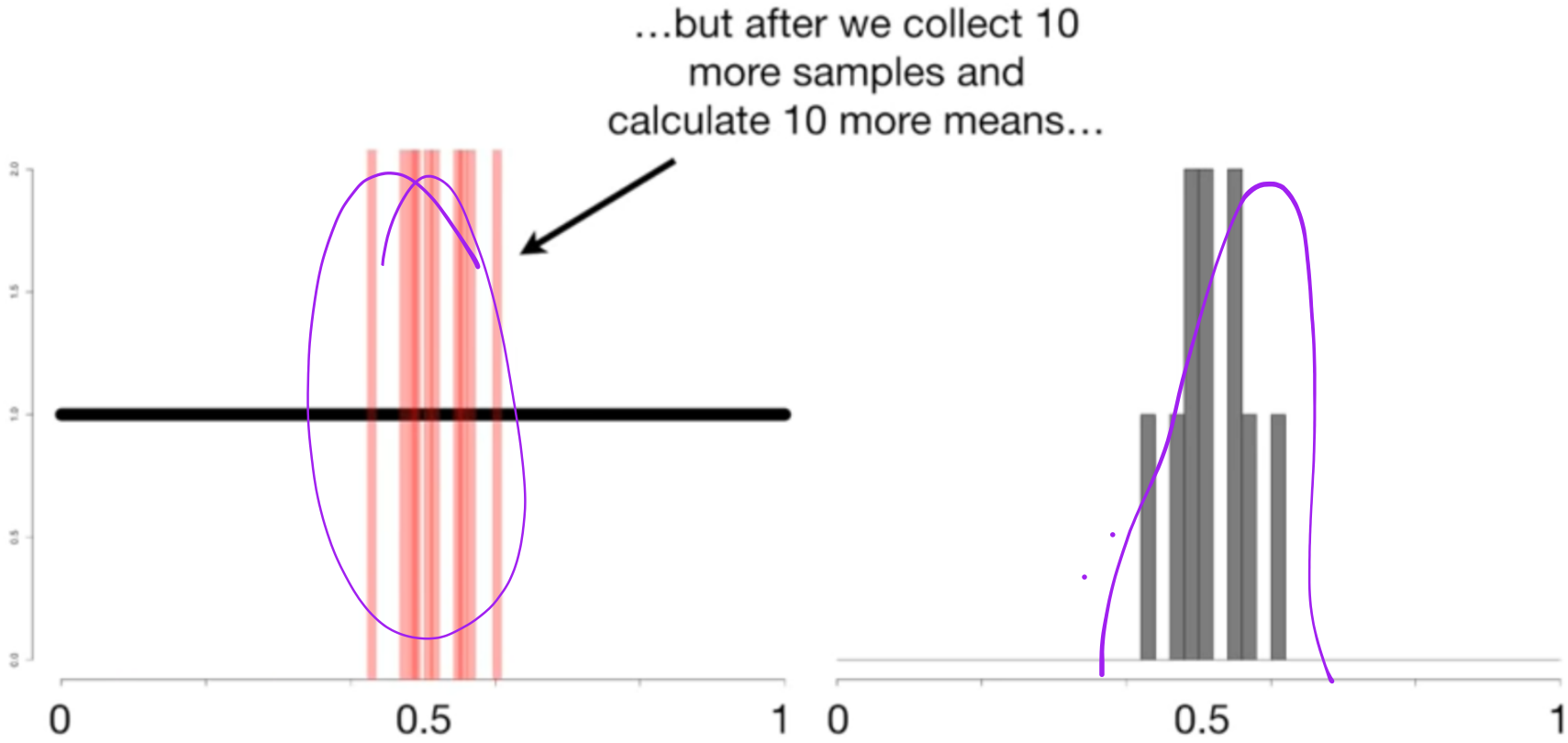
Central Limit Theorem



Central Limit Theorem

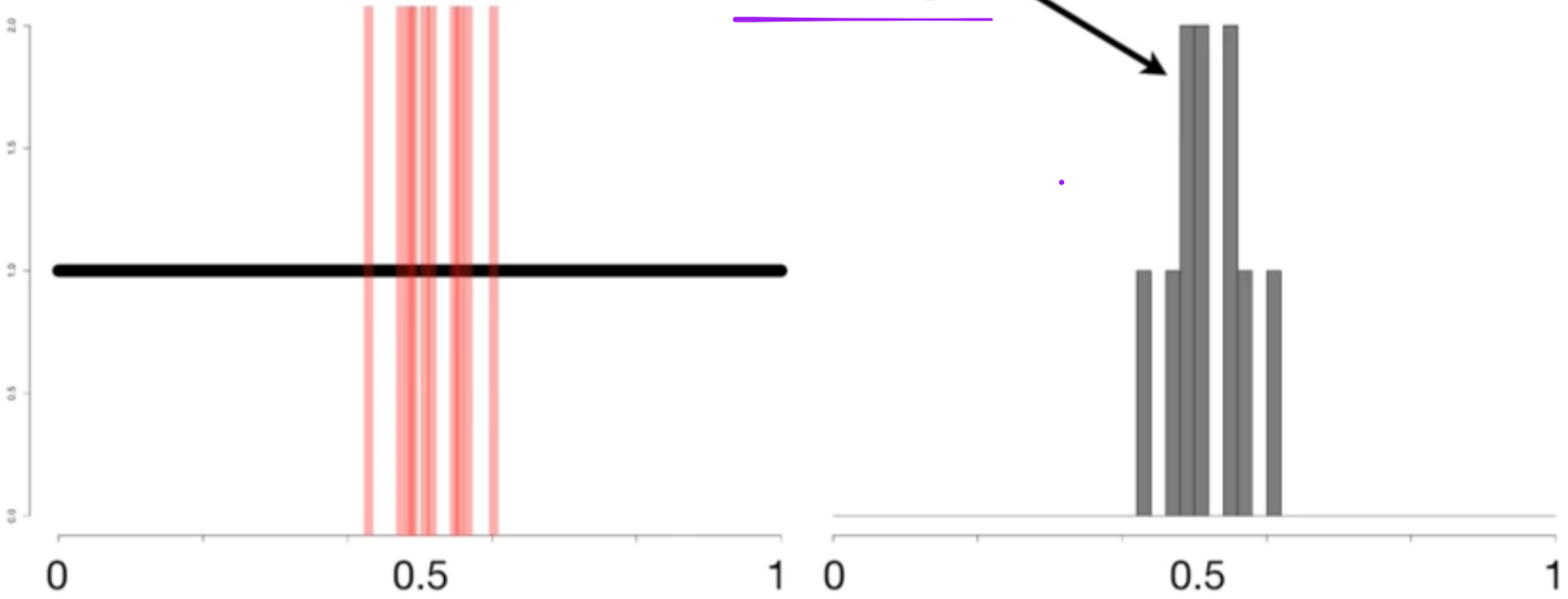


Central Limit Theorem

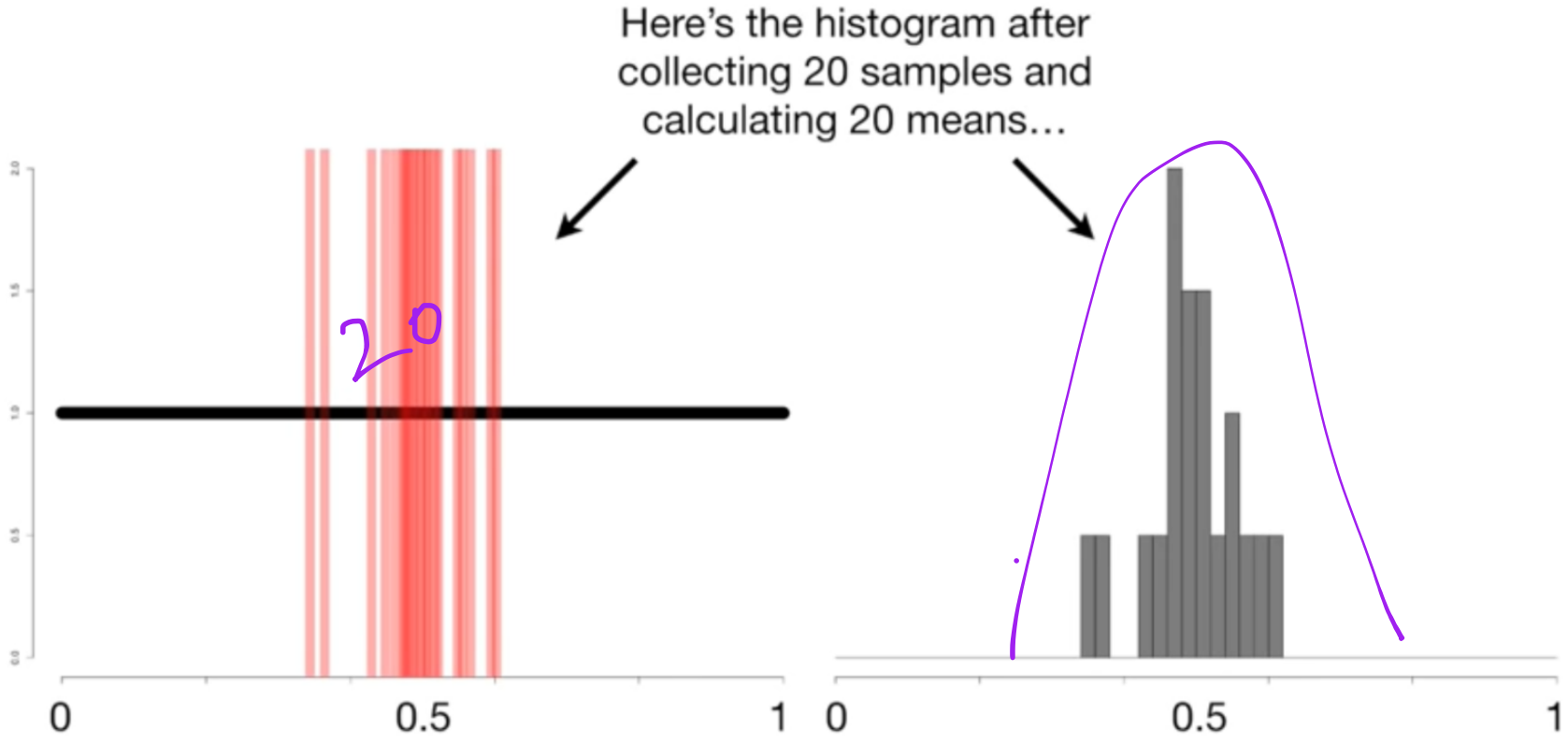


Central Limit Theorem

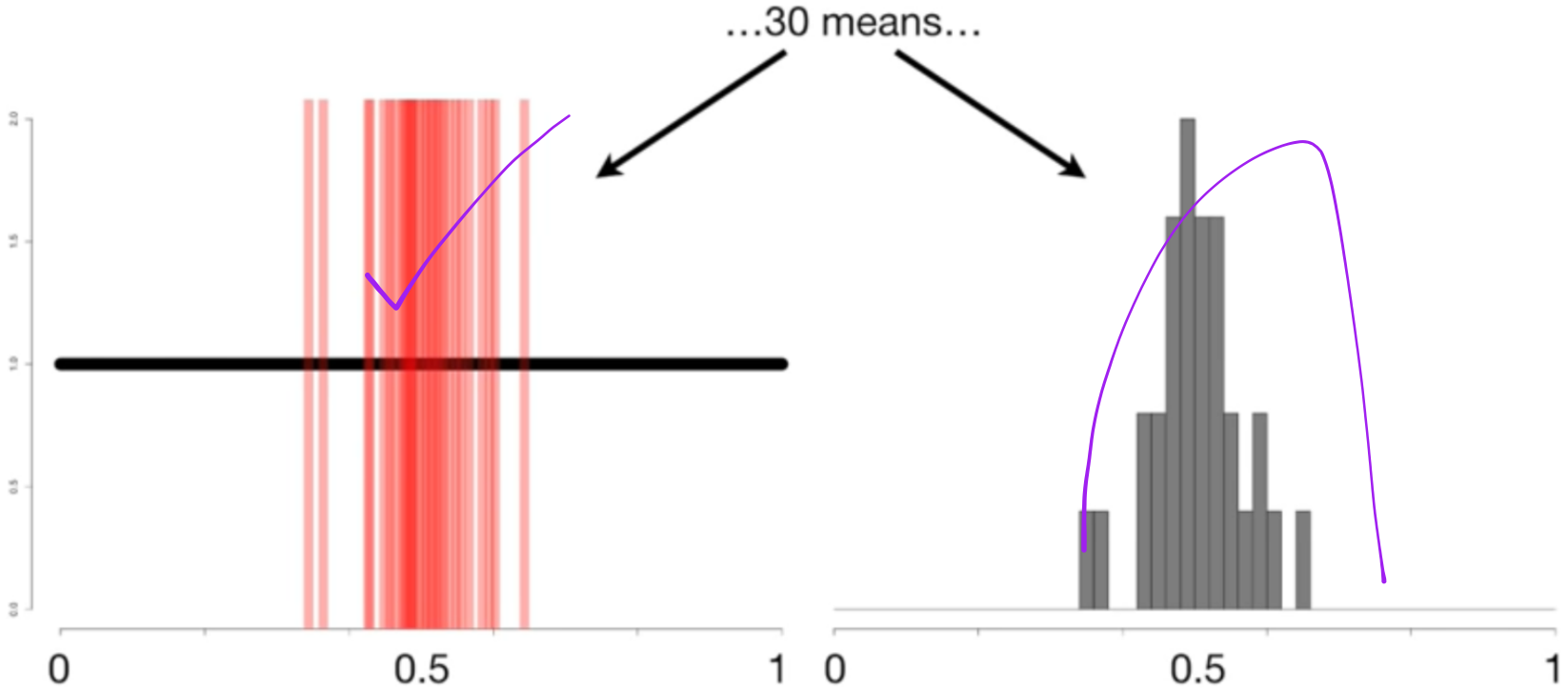
…the histogram starts to look at little more interesting.



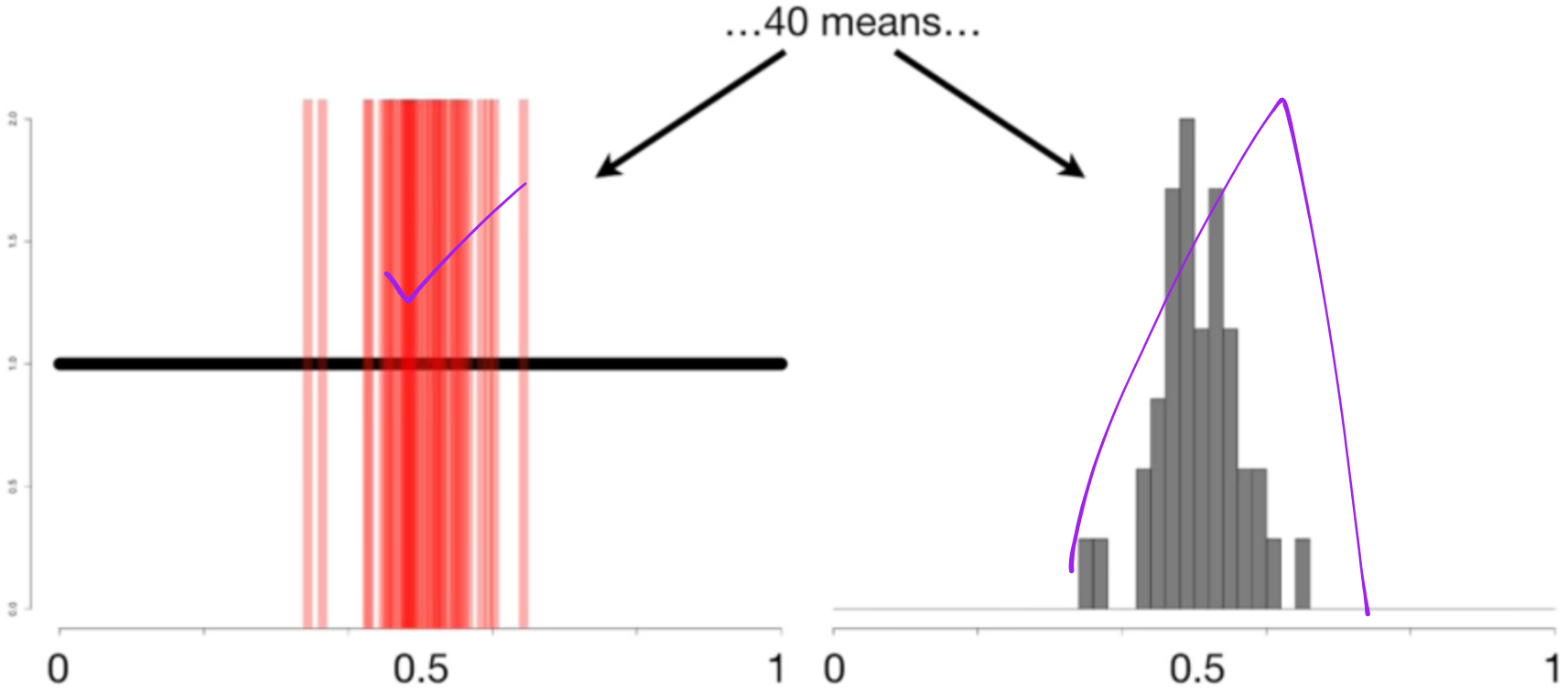
Central Limit Theorem



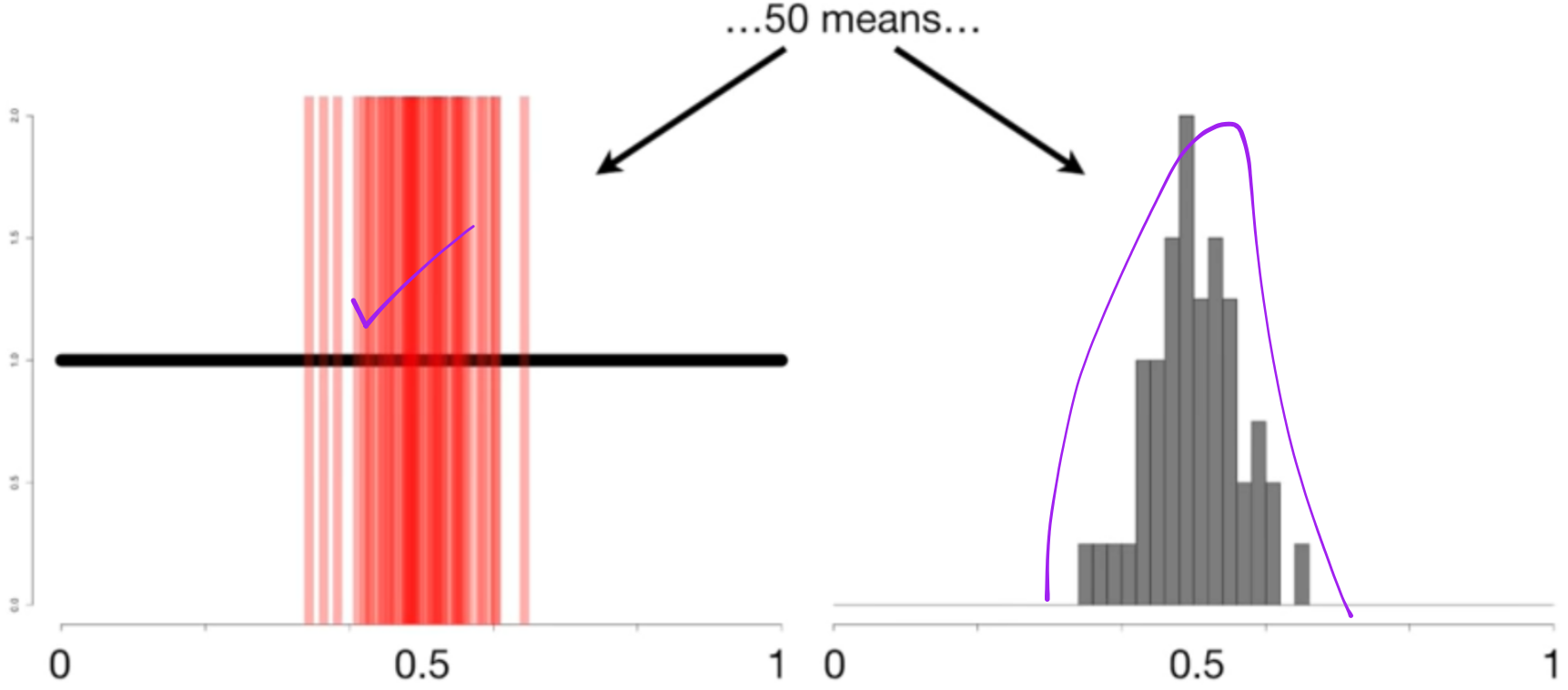
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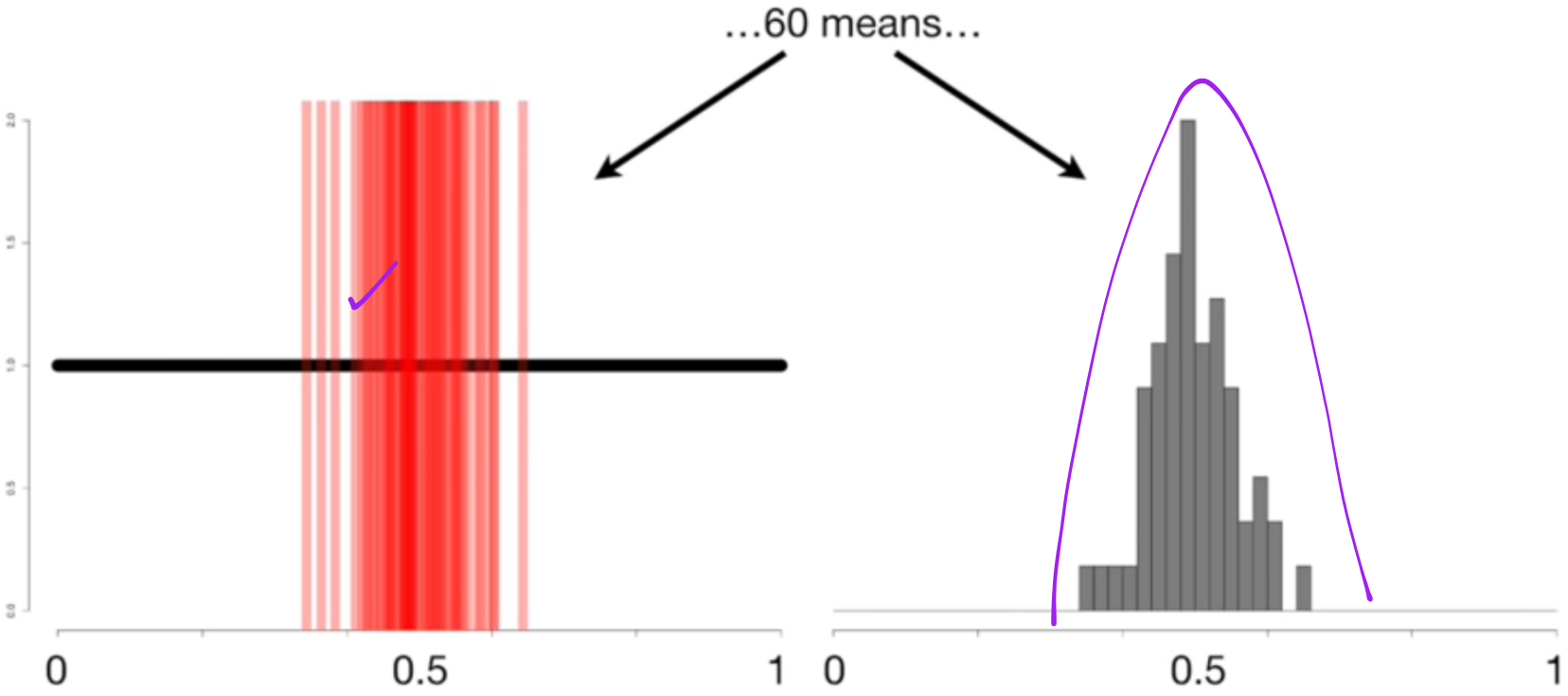
Central Limit Theorem



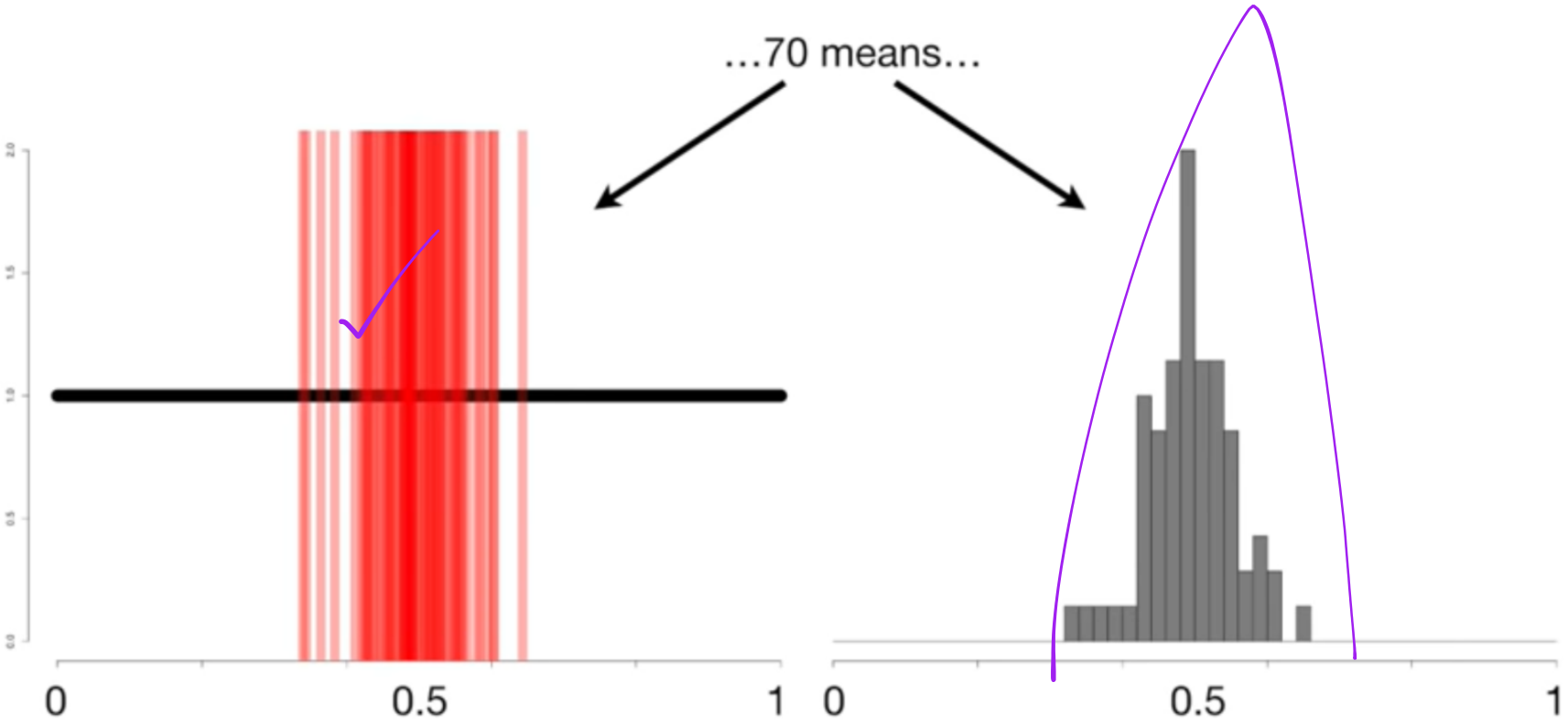
Central Limit Theorem



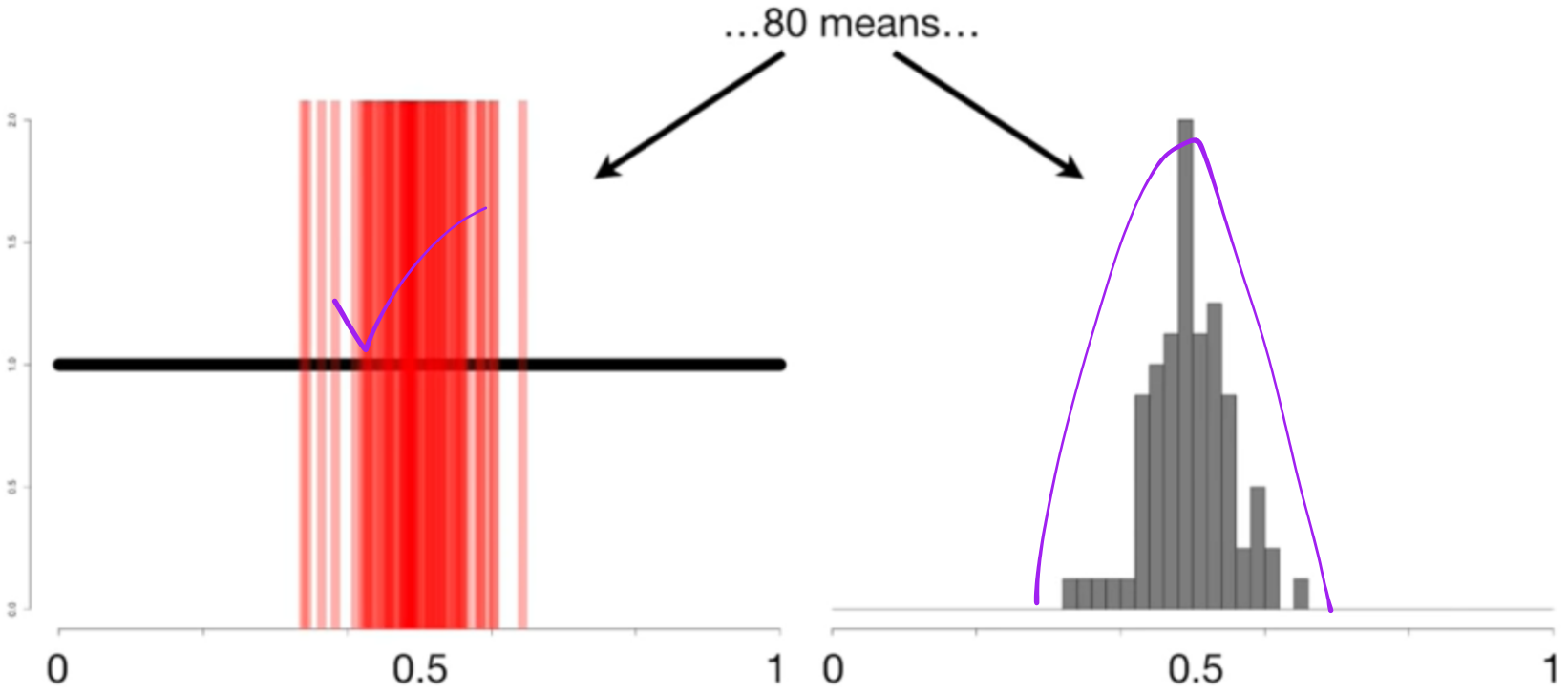
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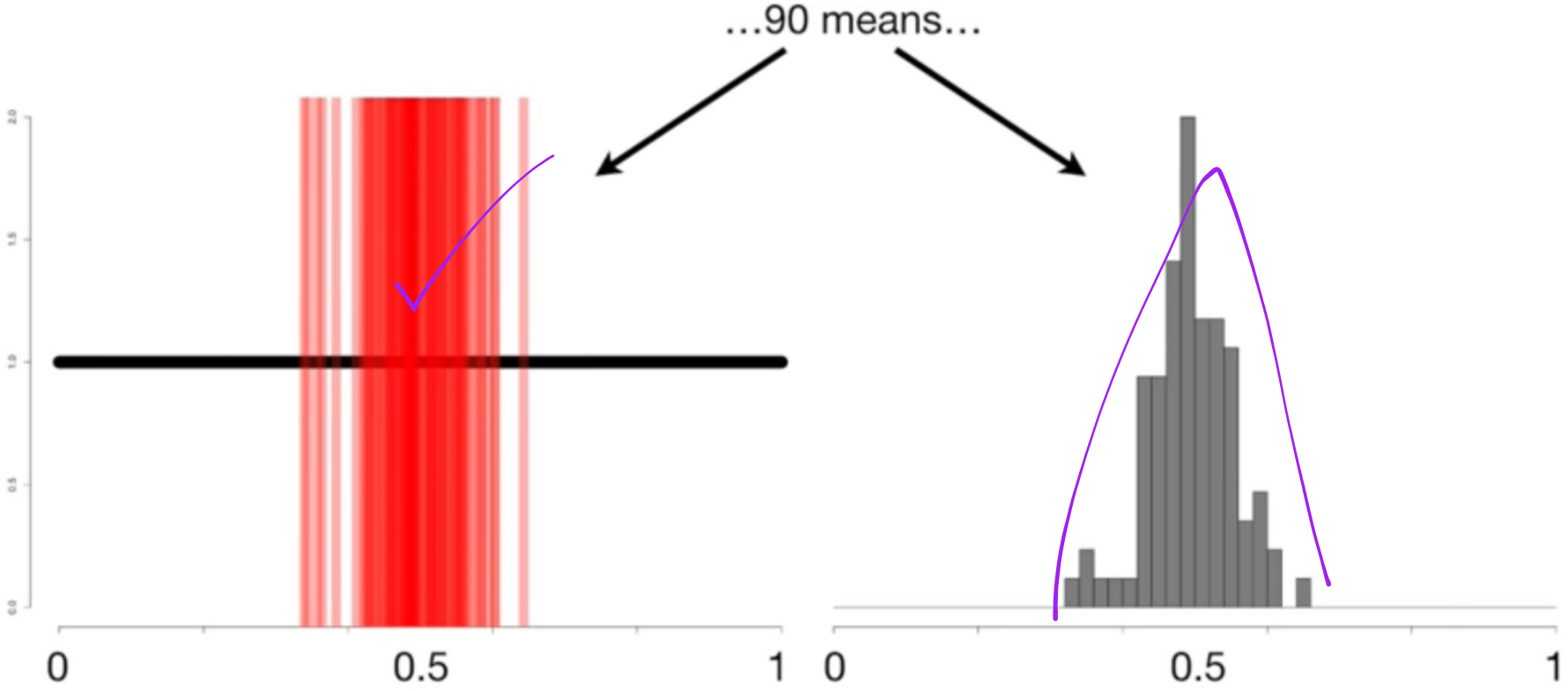
Central Limit Theorem



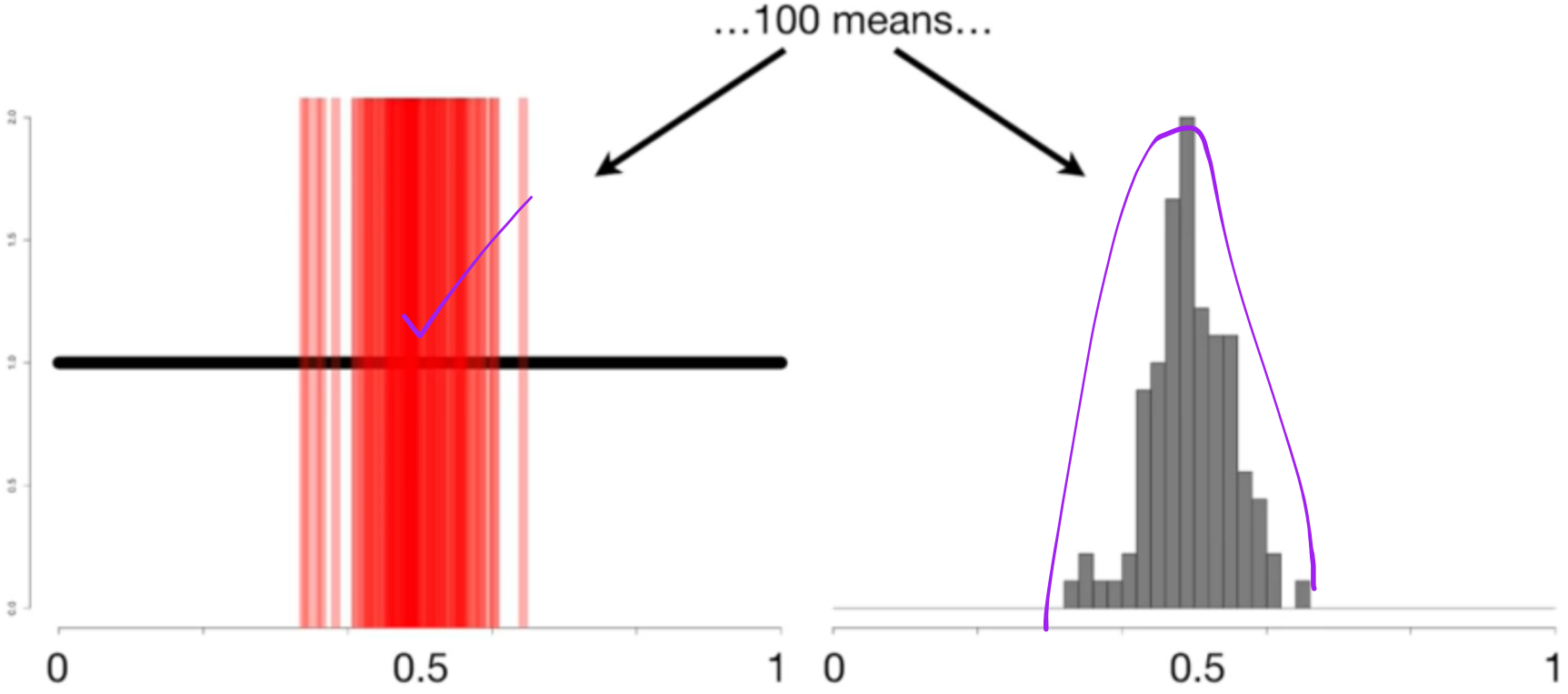
Central Limit Theorem



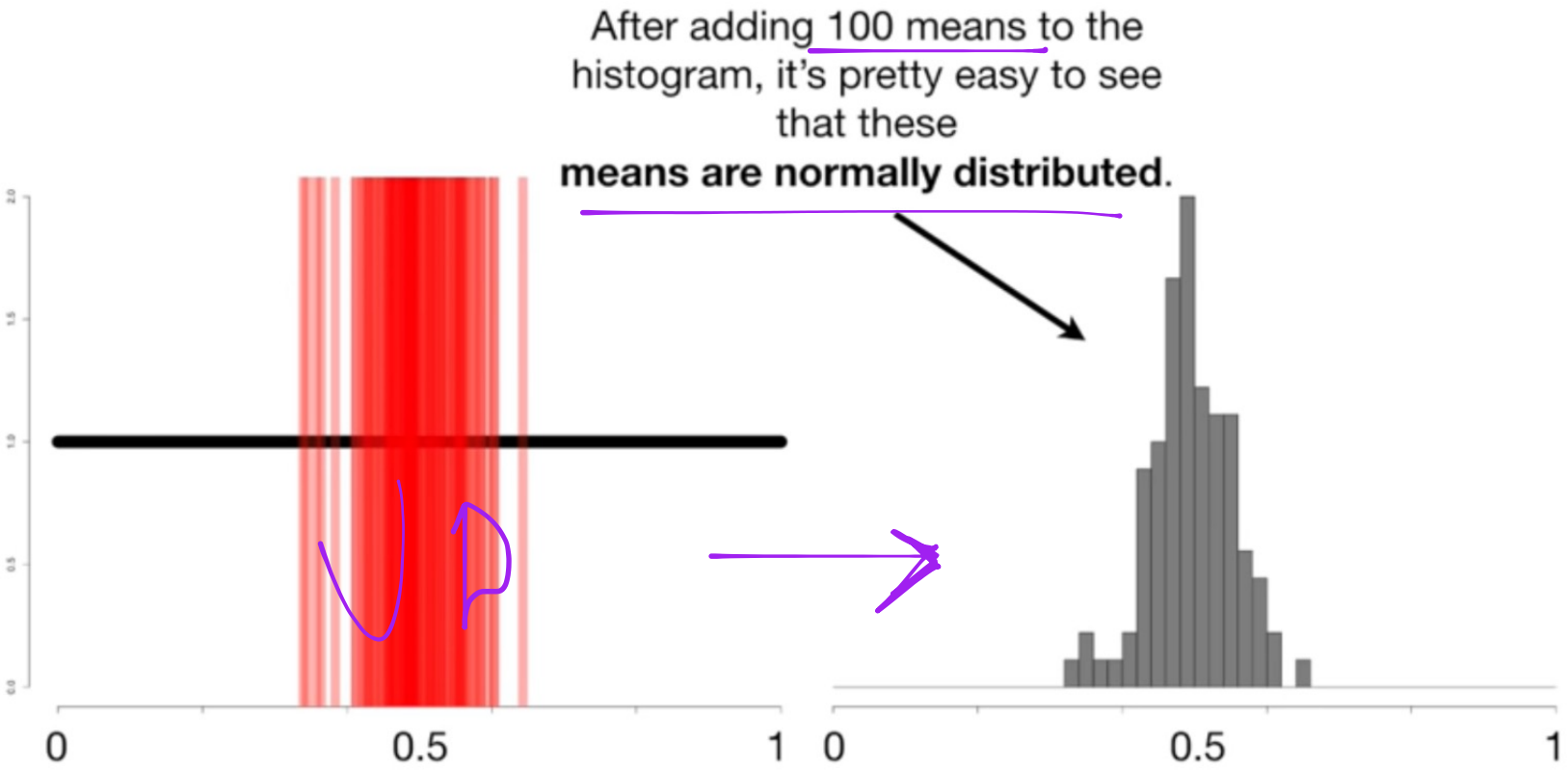
Central Limit Theorem



Central Limit Theorem

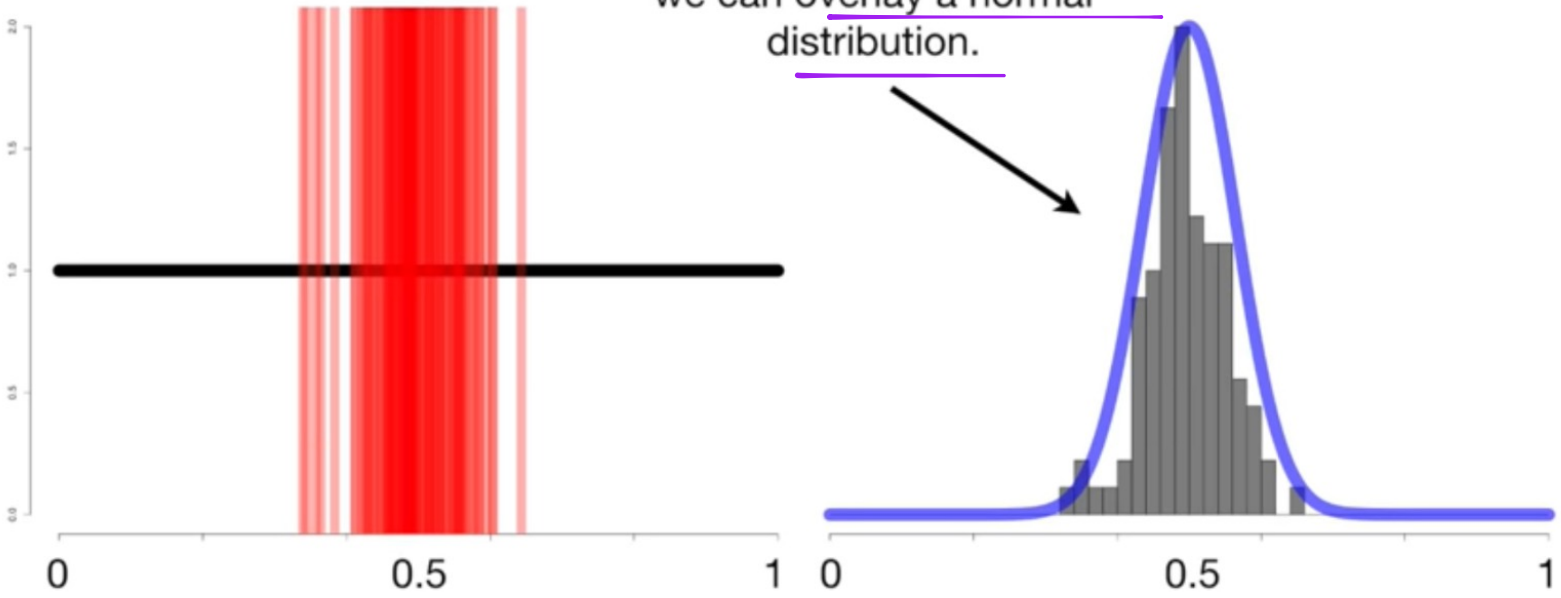


Central Limit Theorem

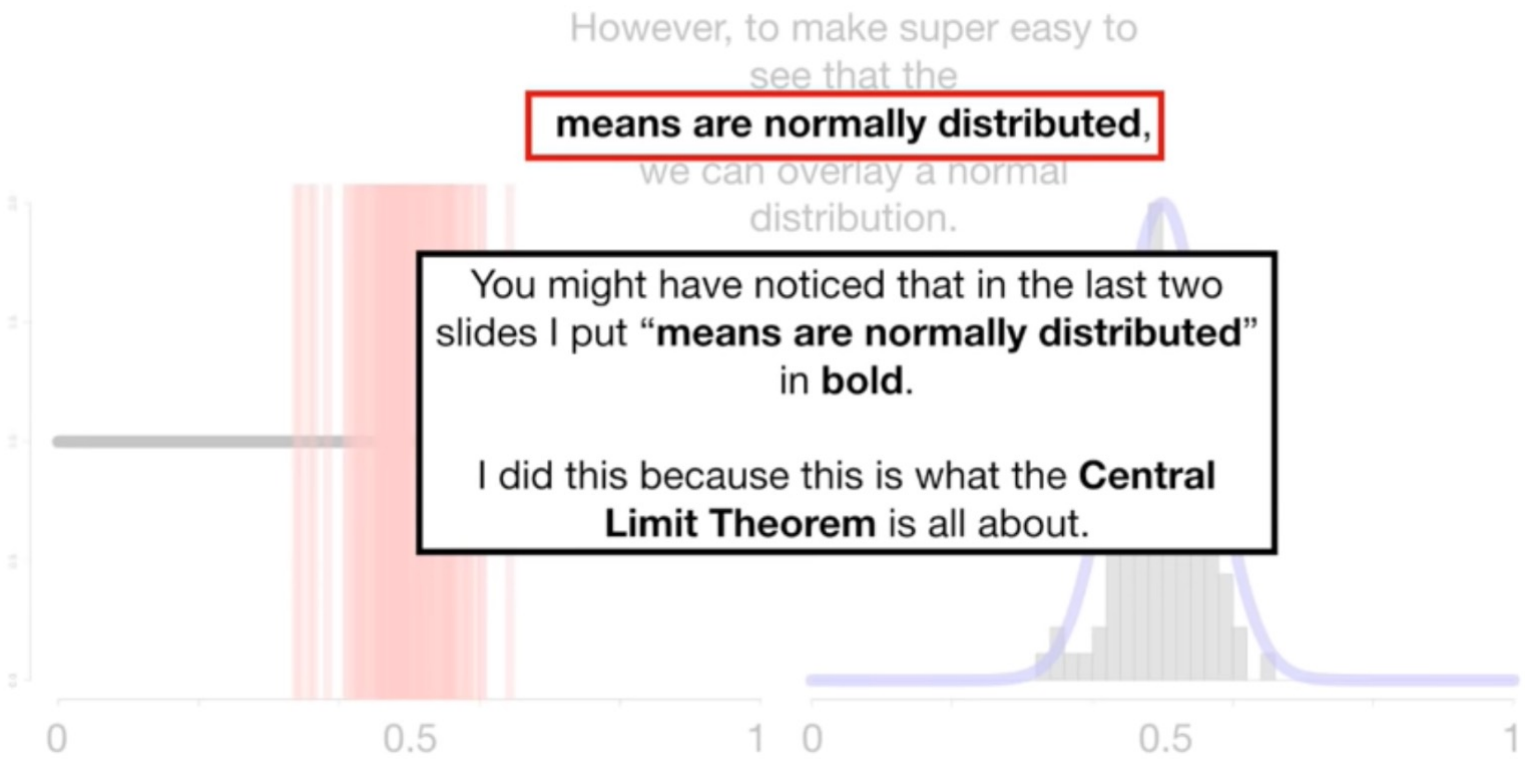


Central Limit Theorem

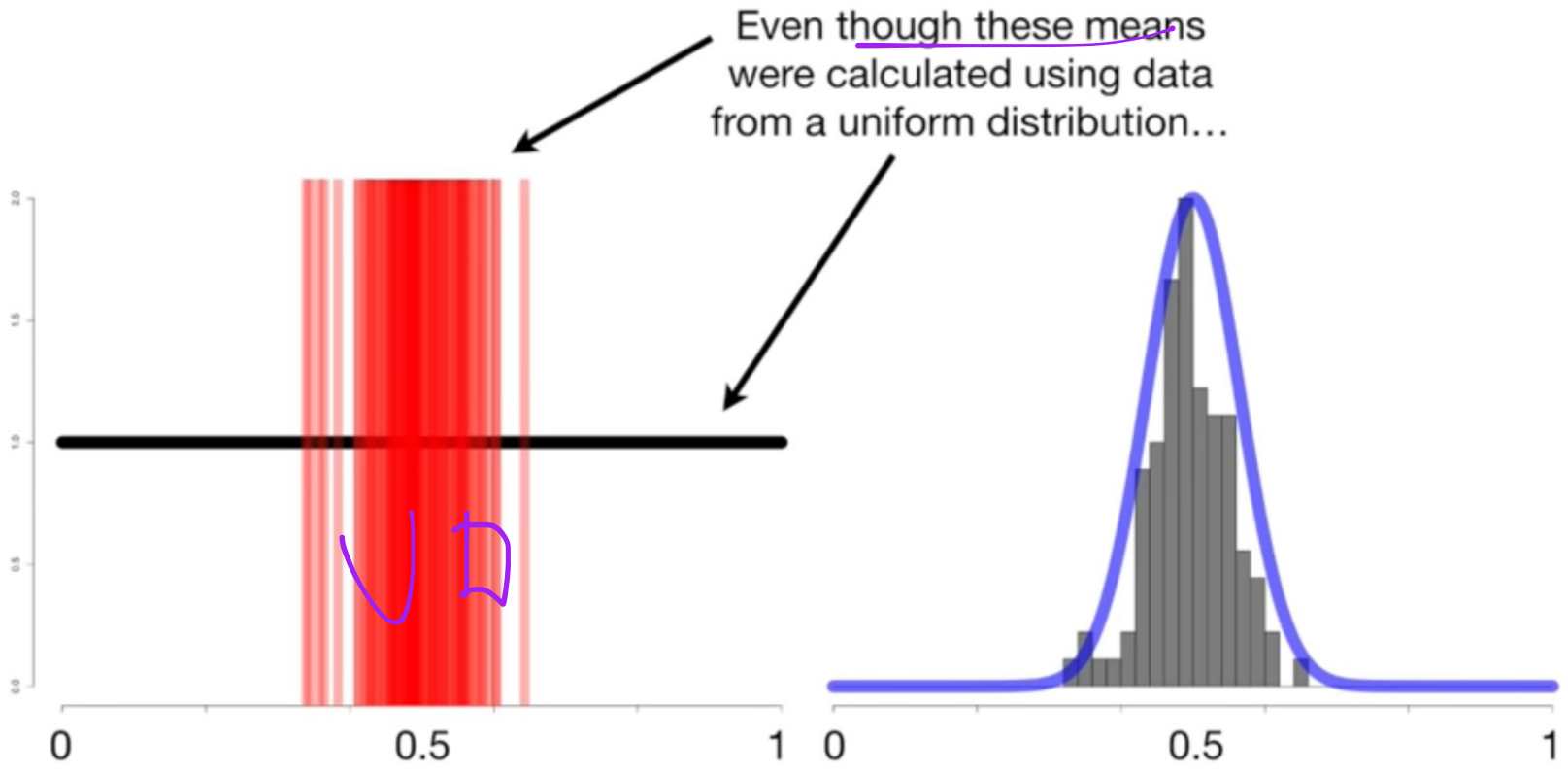
However, to make super easy to see that the **means are normally distributed**, we can overlay a normal distribution.



Central Limit Theorem

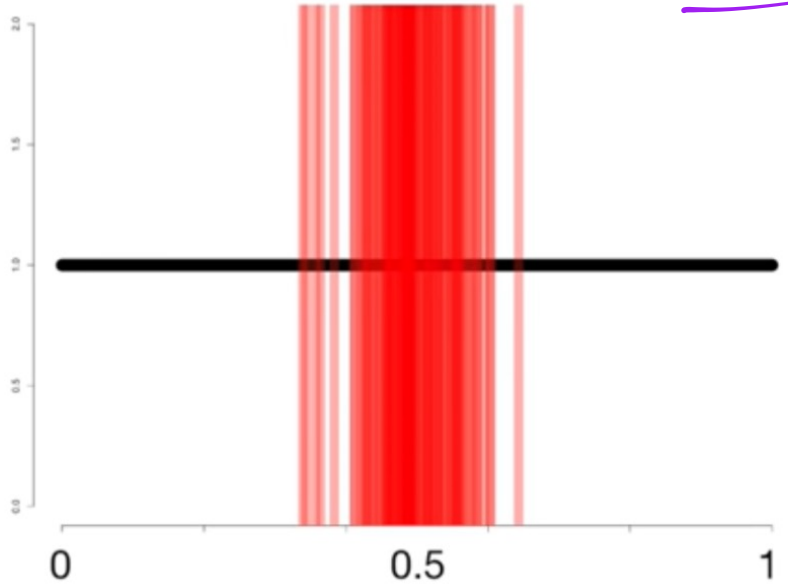


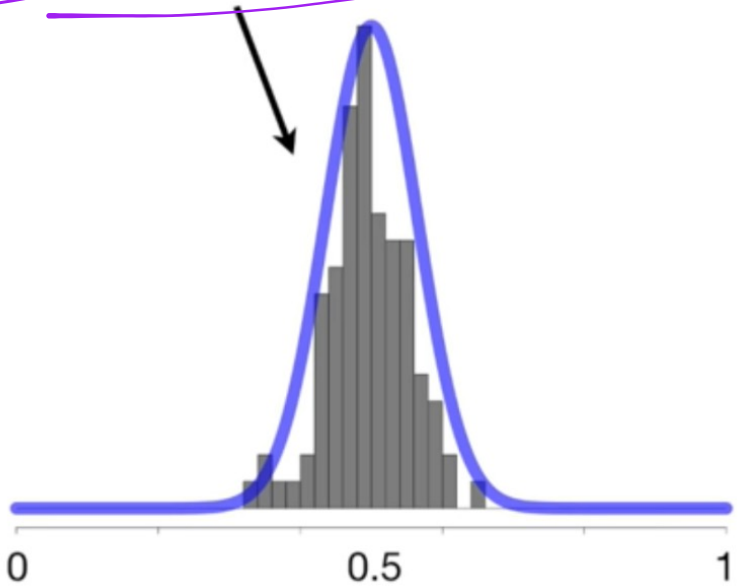
Central Limit Theorem



Central Limit Theorem

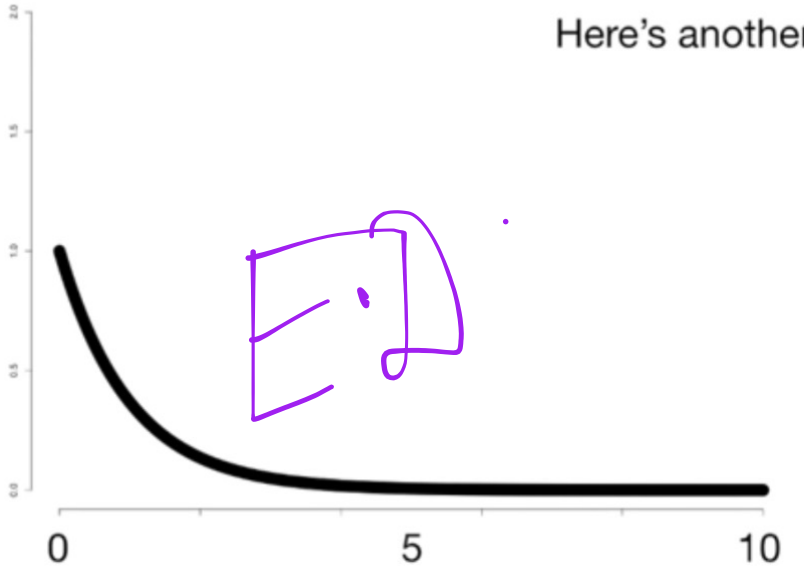
…the means themselves are not uniformly distributed. Instead, the **means are normally distributed**.



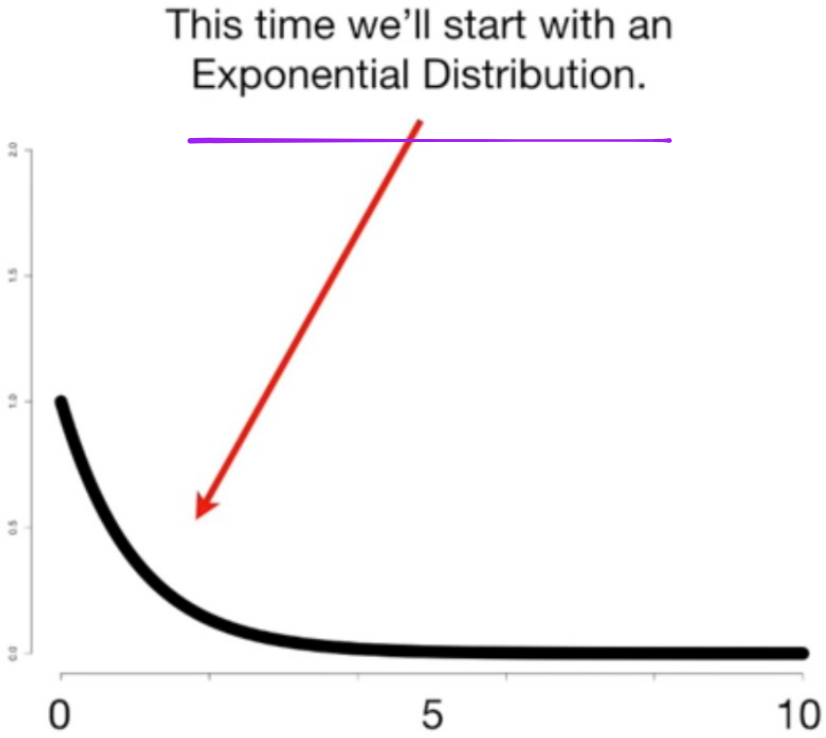


Central Limit Theorem

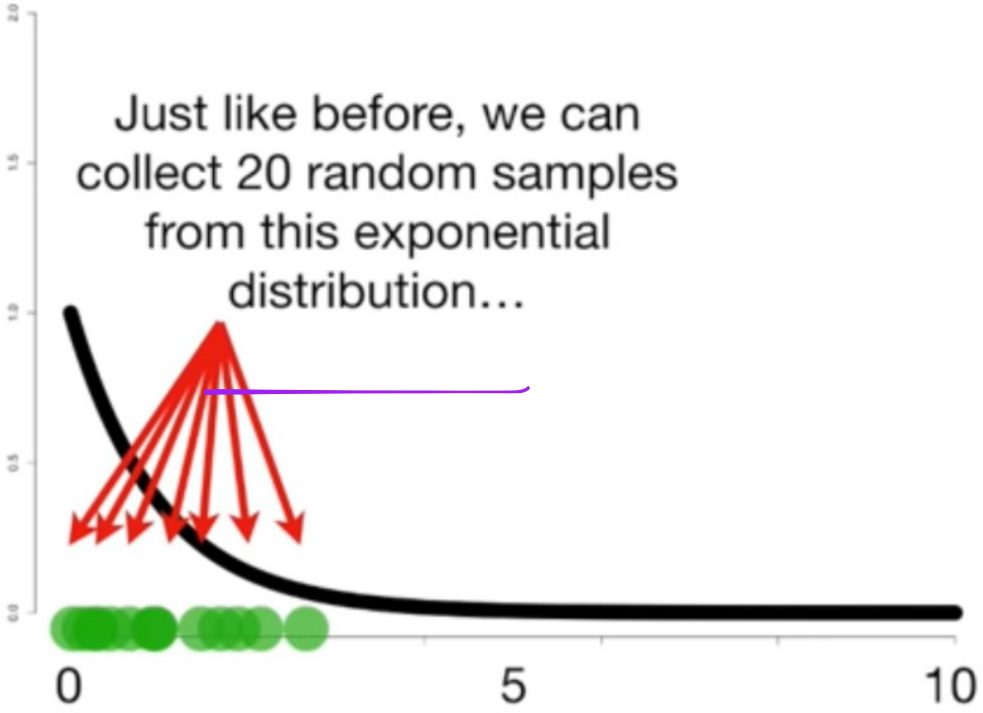
Here's another example…



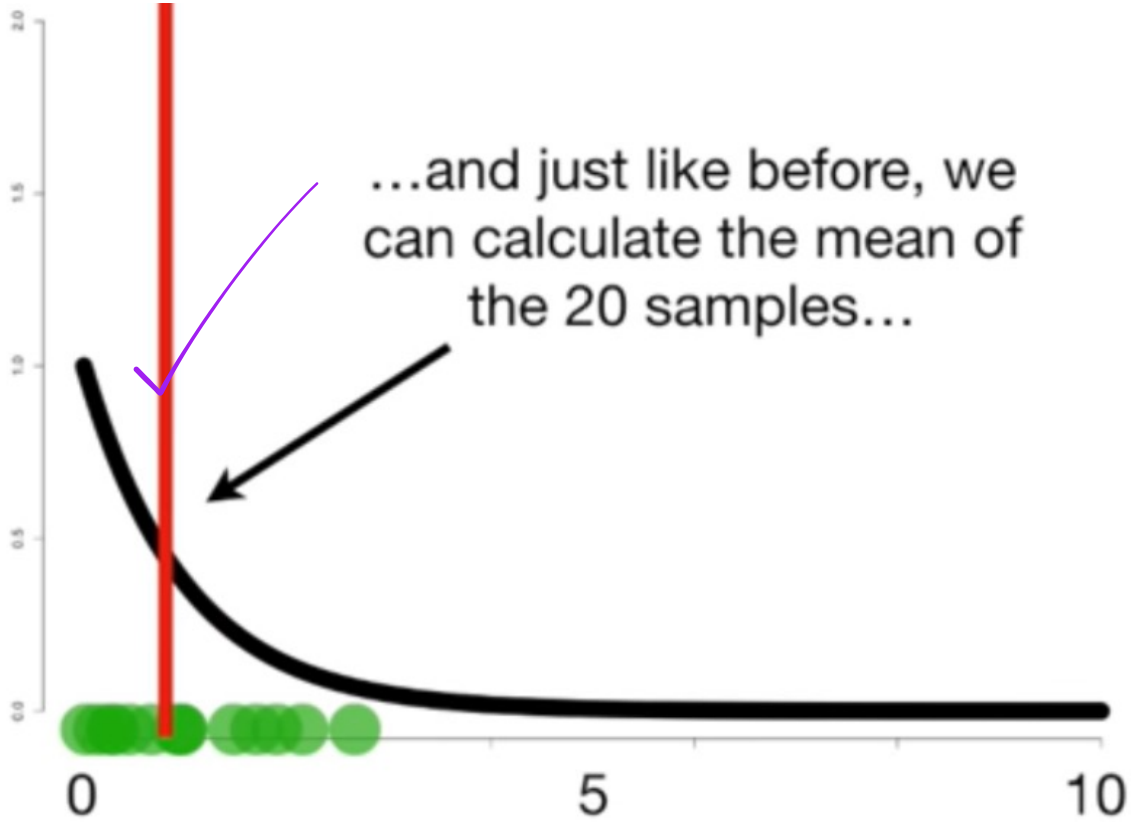
Central Limit Theorem



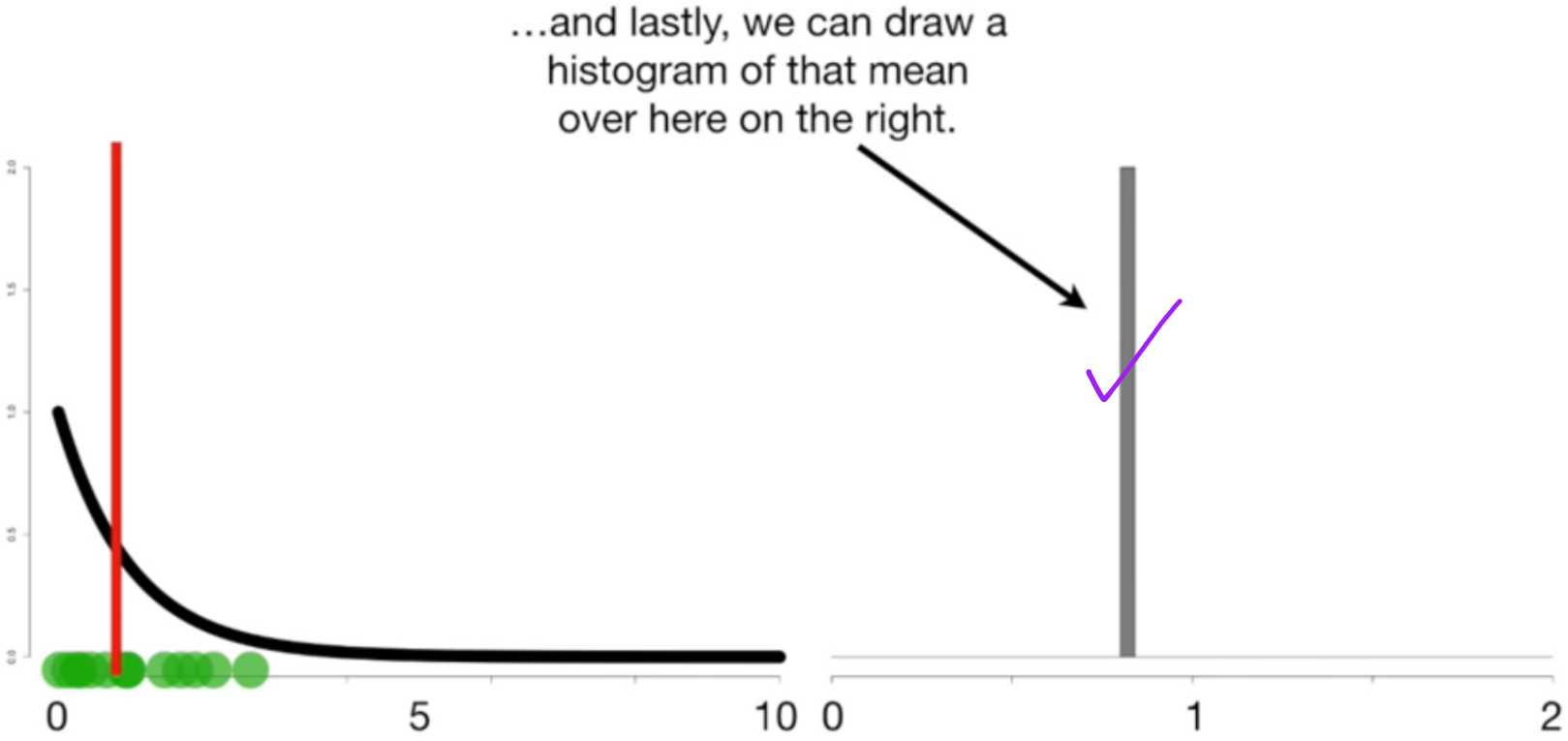
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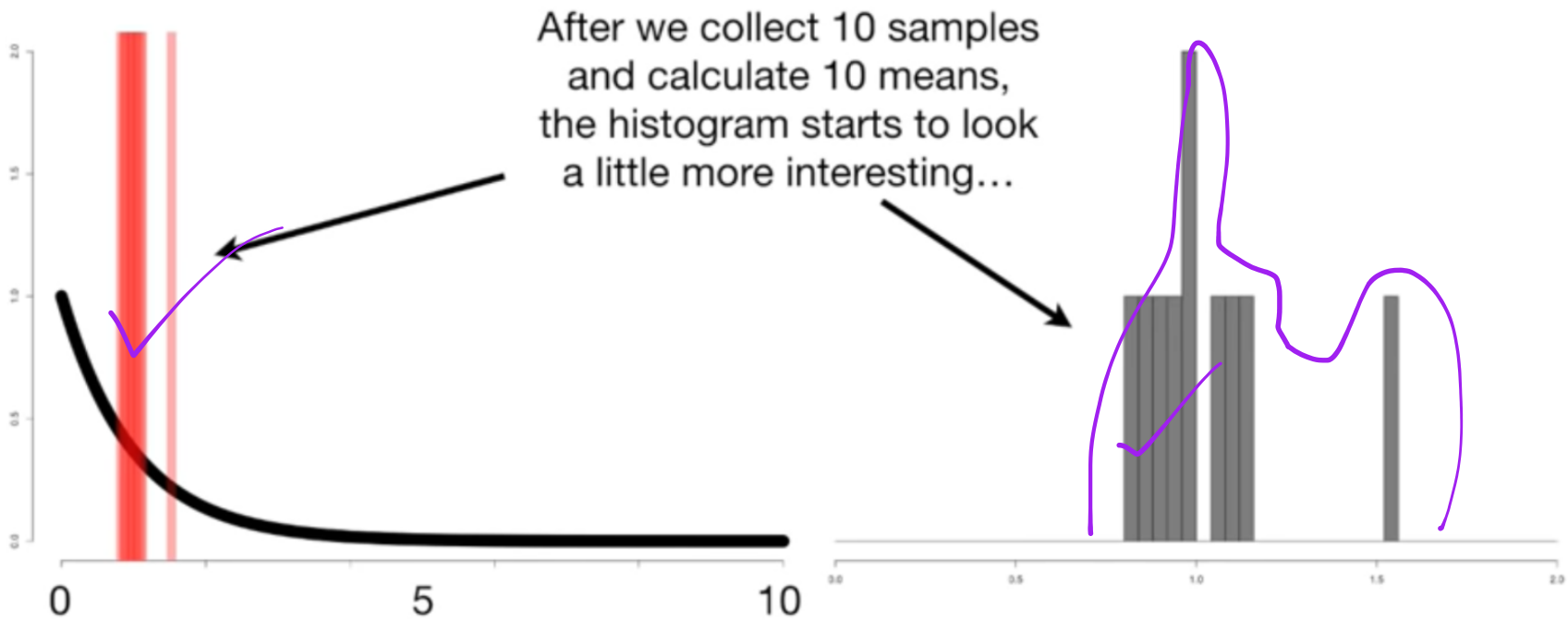
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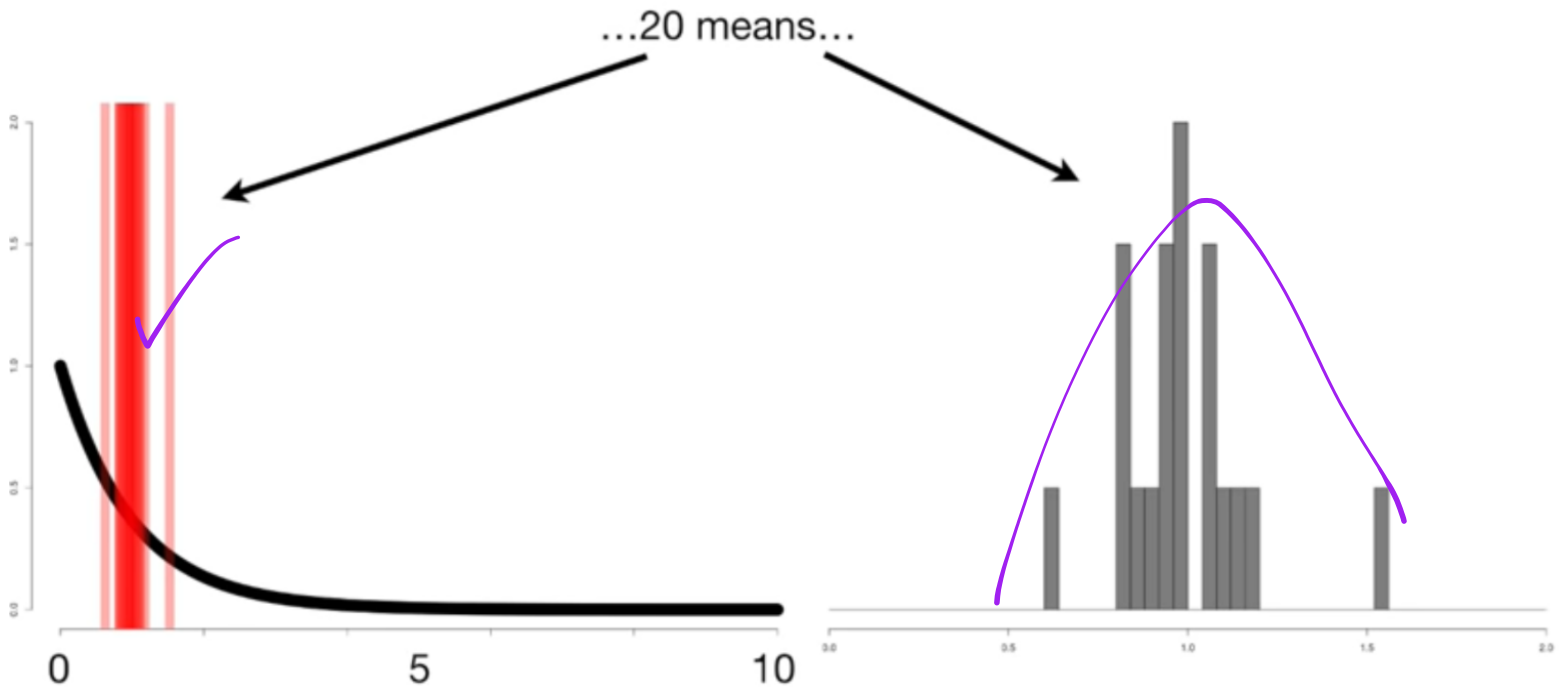
Central Limit Theorem



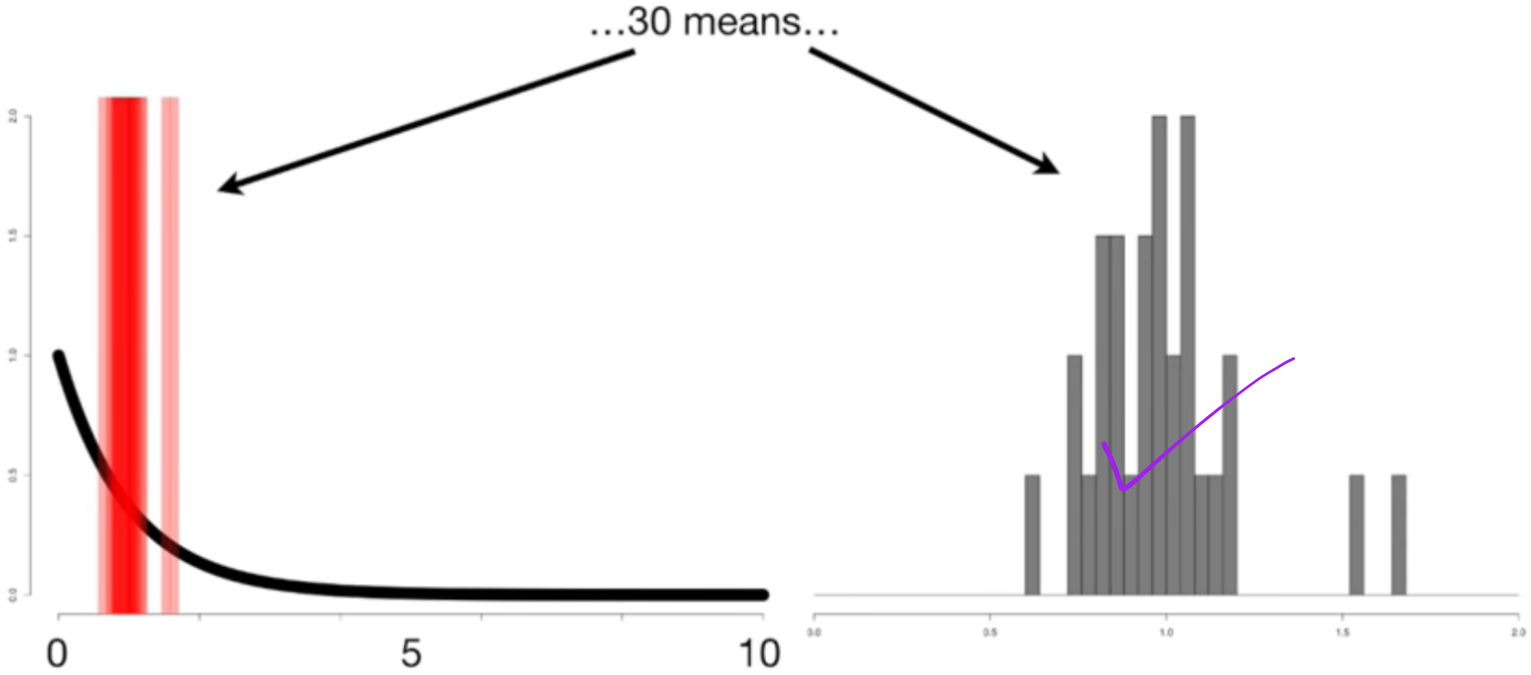
Central Limit Theorem



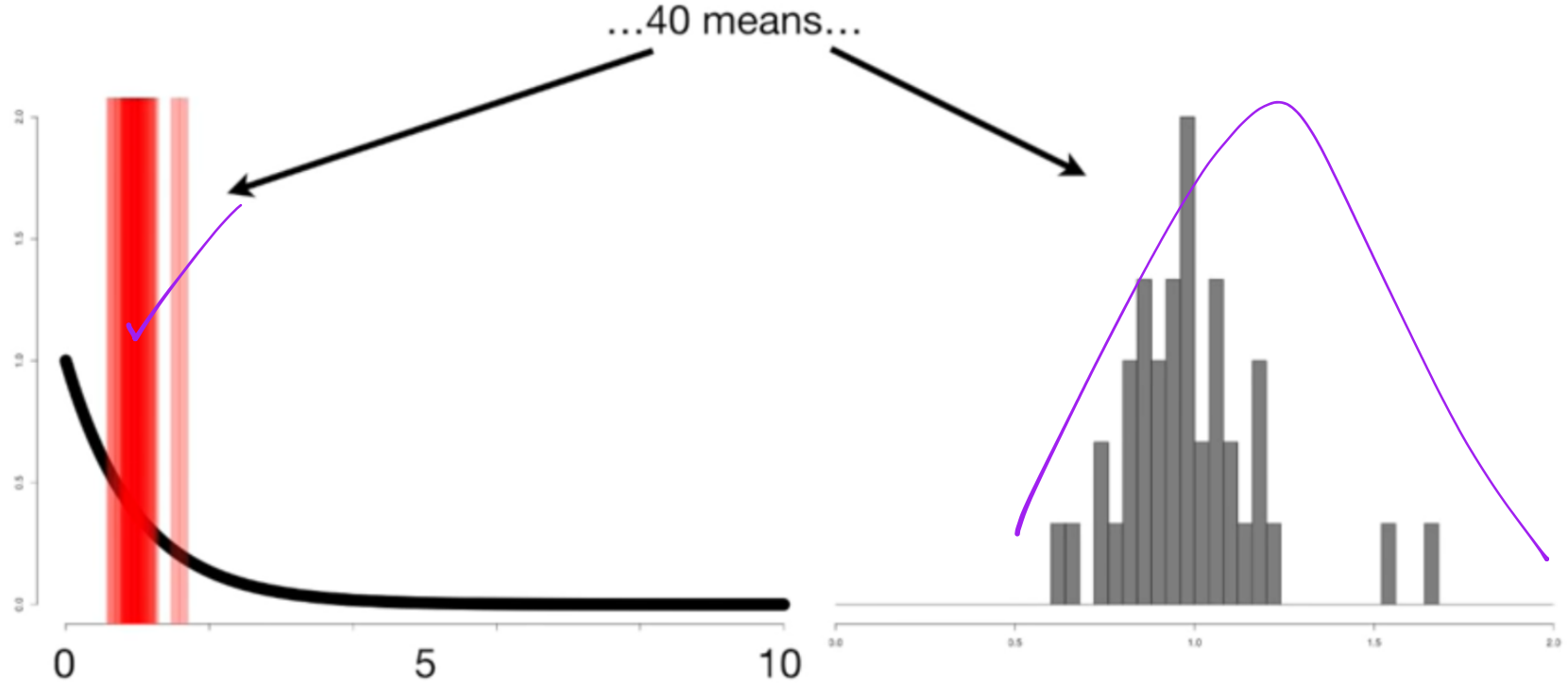
Central Limit Theorem



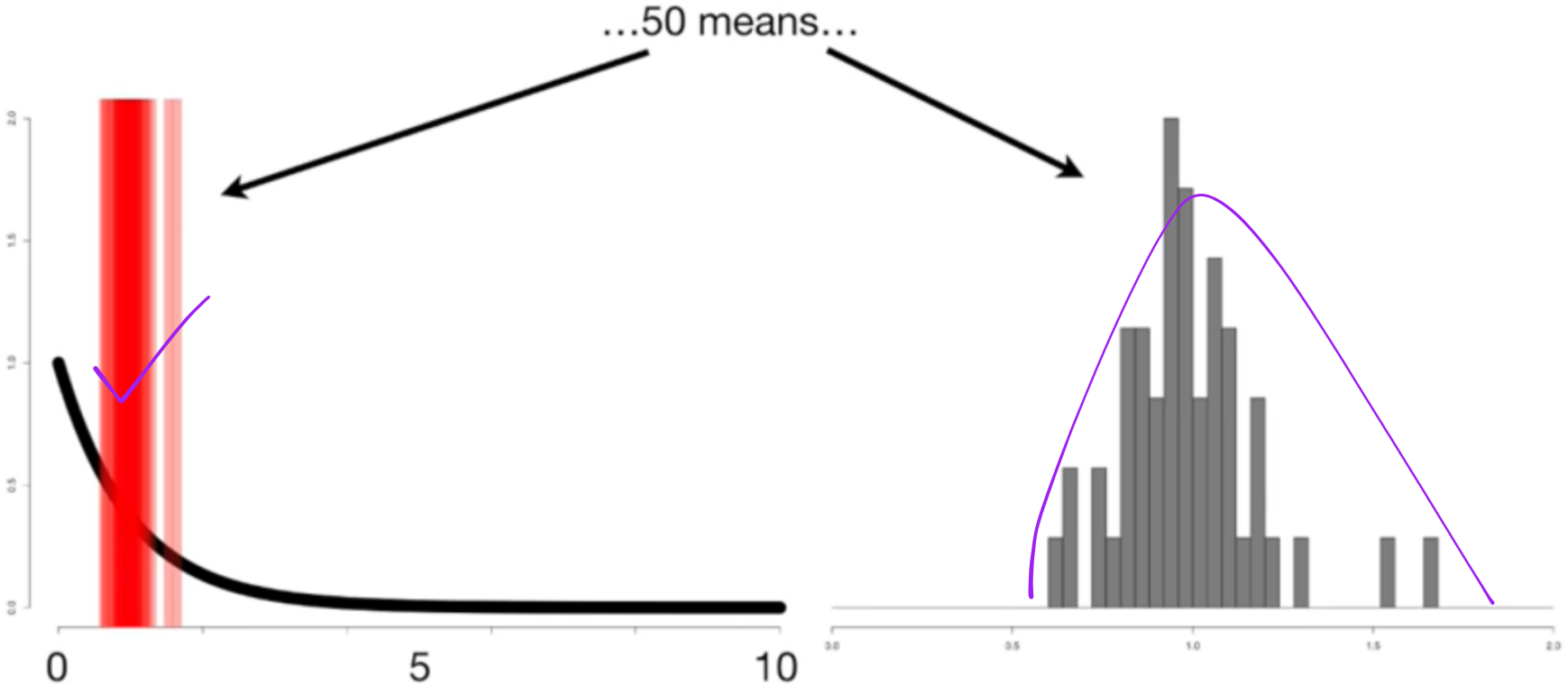
Central Limit Theorem



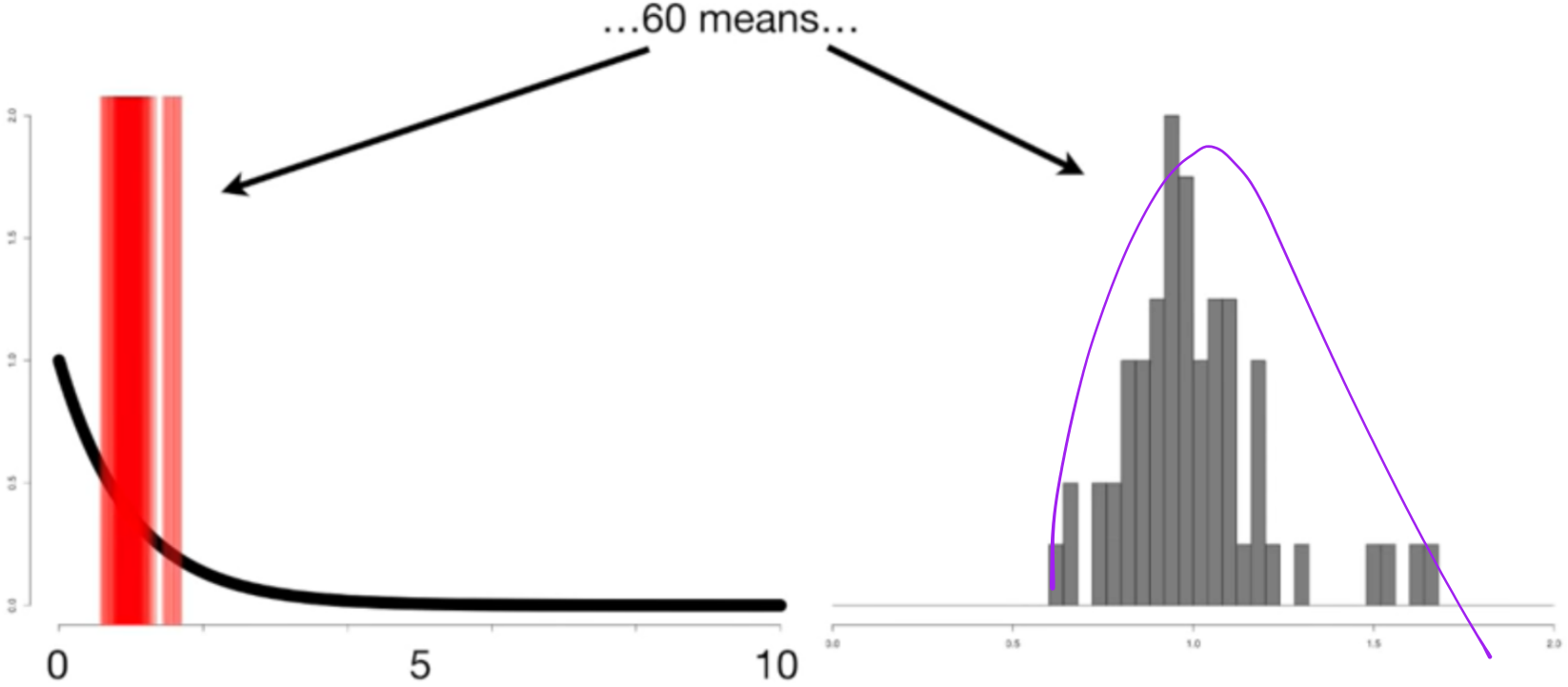
Central Limit Theorem



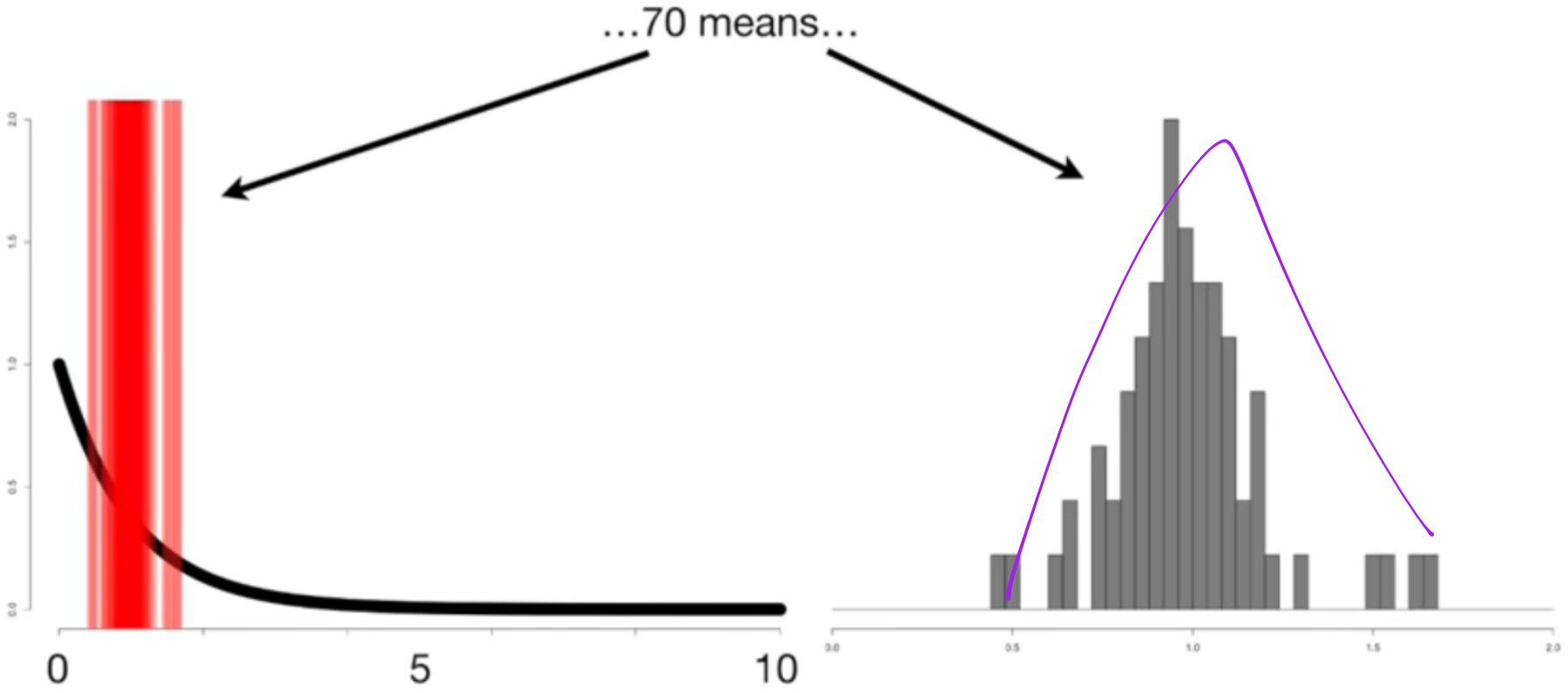
Central Limit Theorem



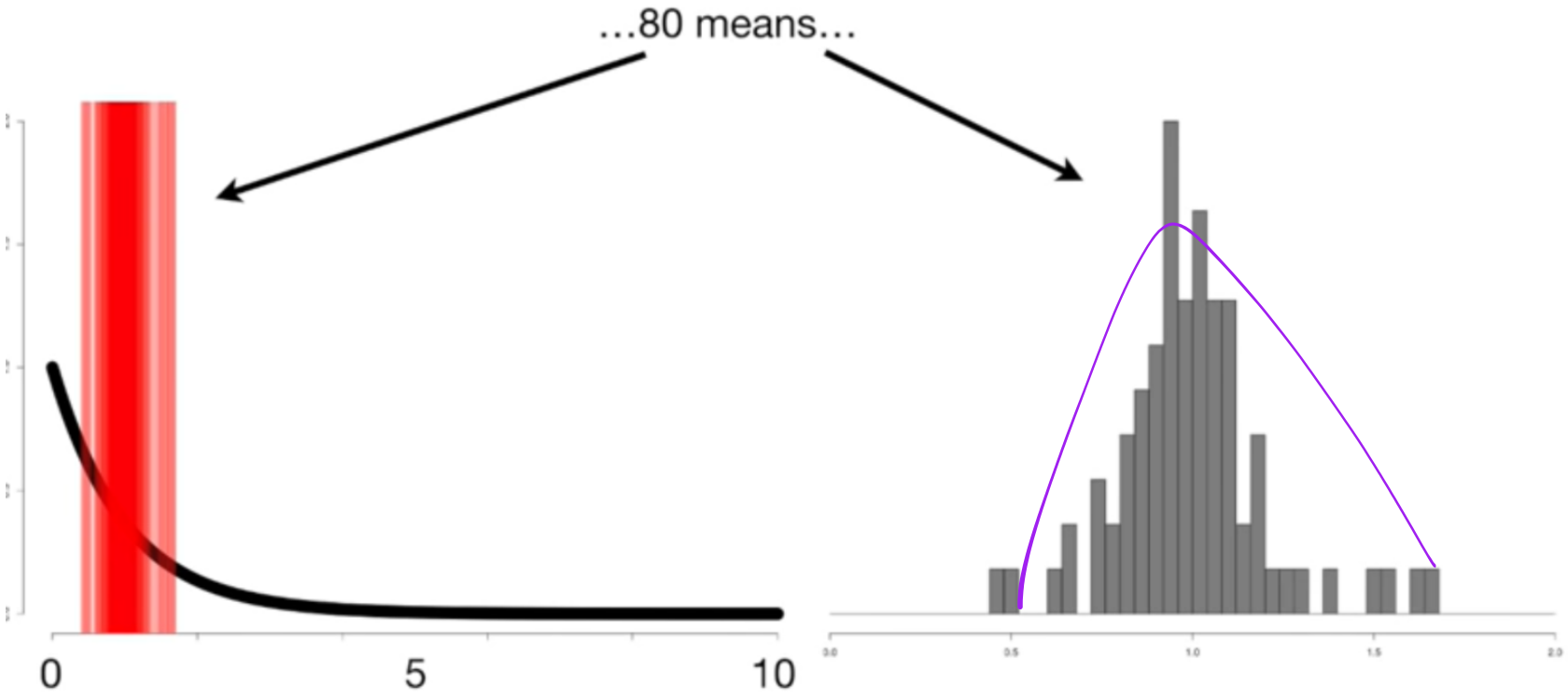
Central Limit Theorem



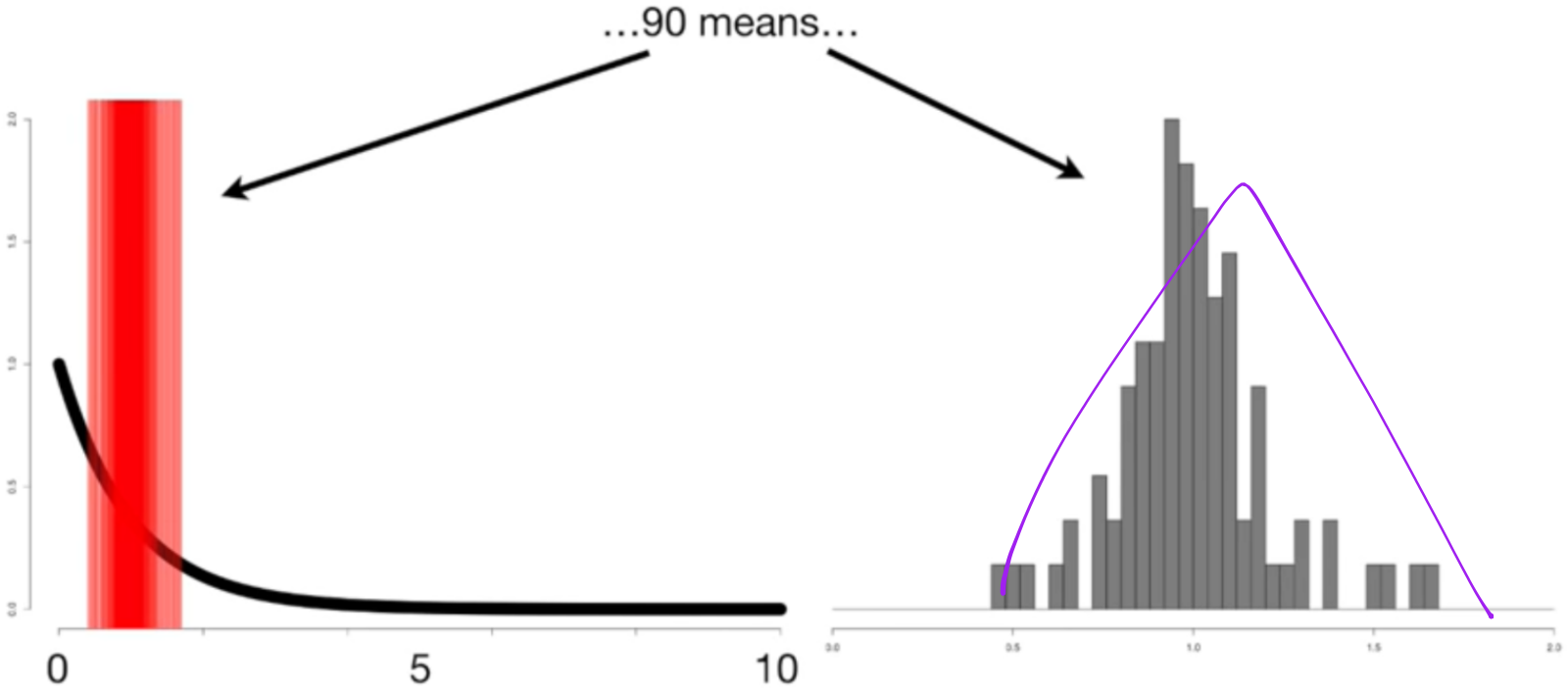
Central Limit Theorem



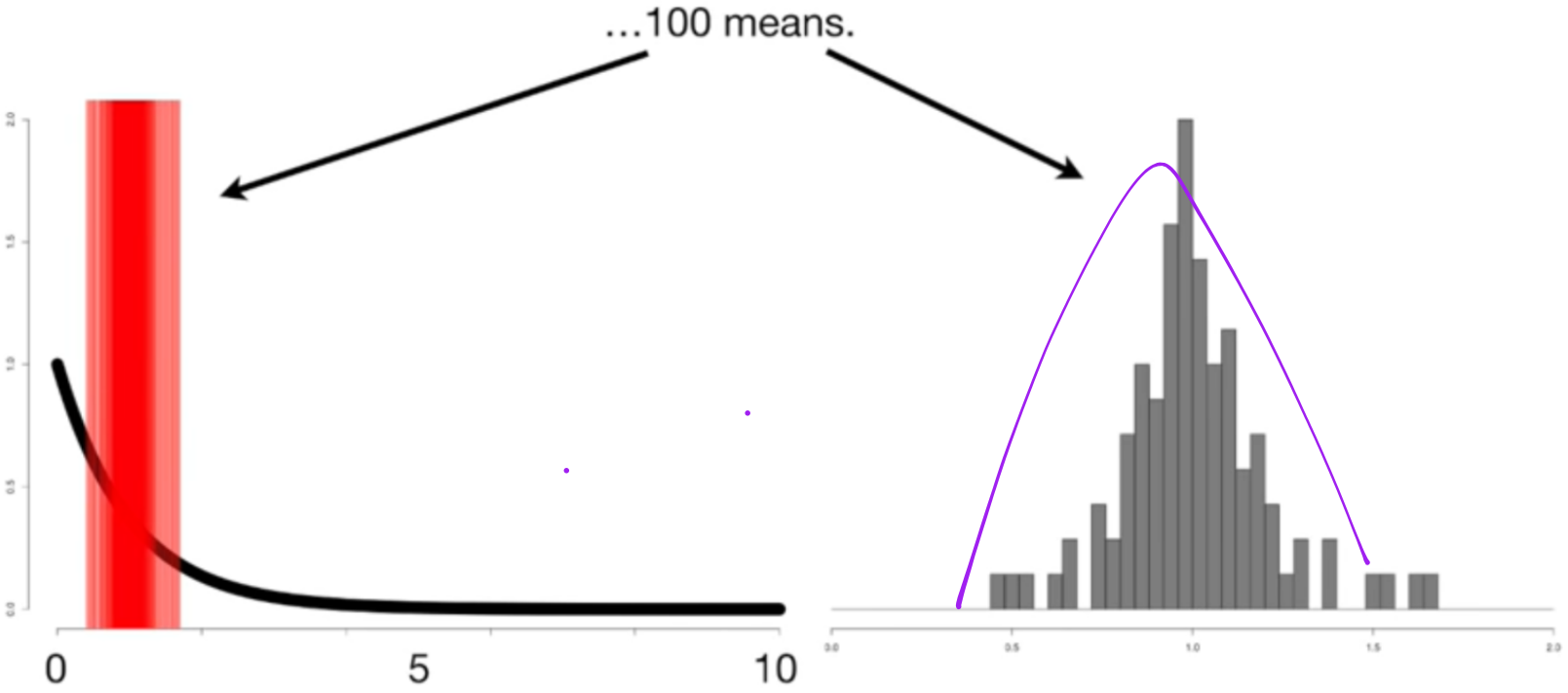
Central Limit Theorem



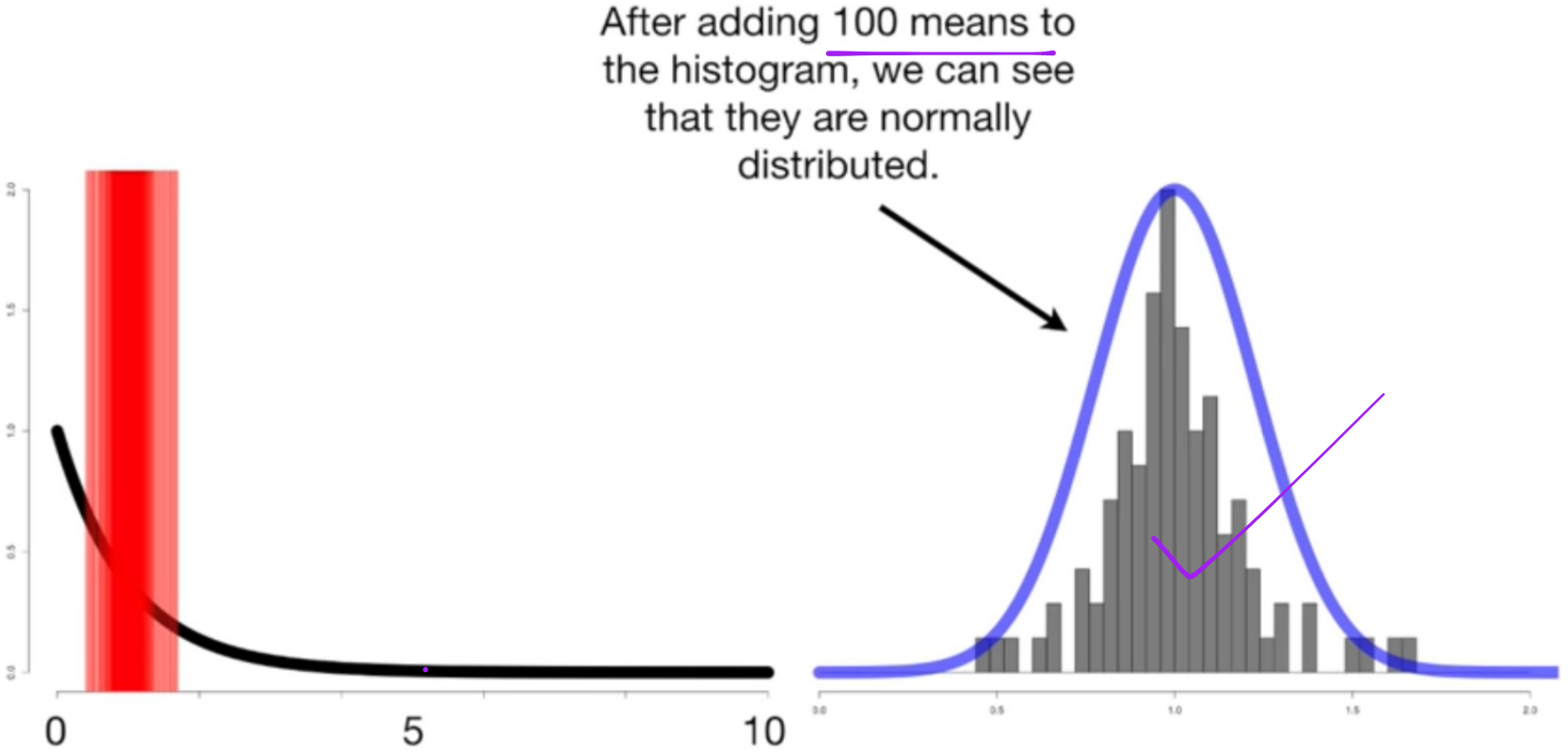
Central Limit Theorem



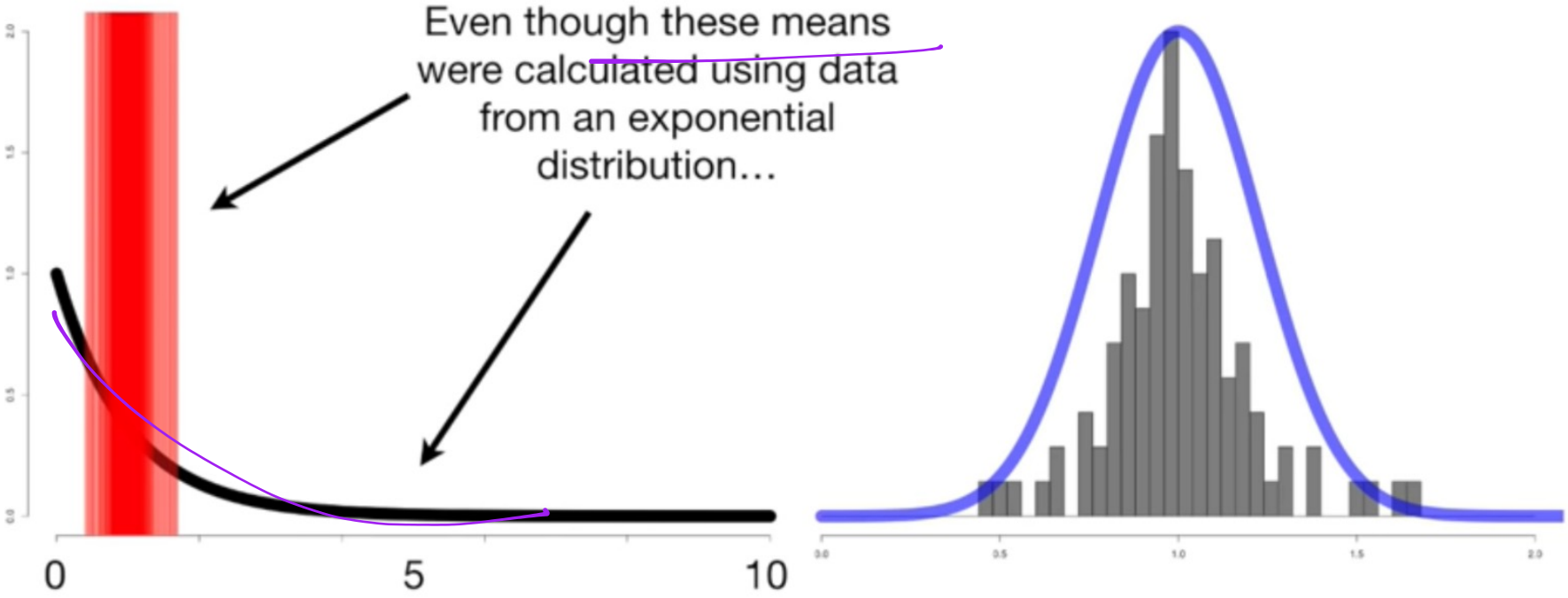
Central Limit Theorem



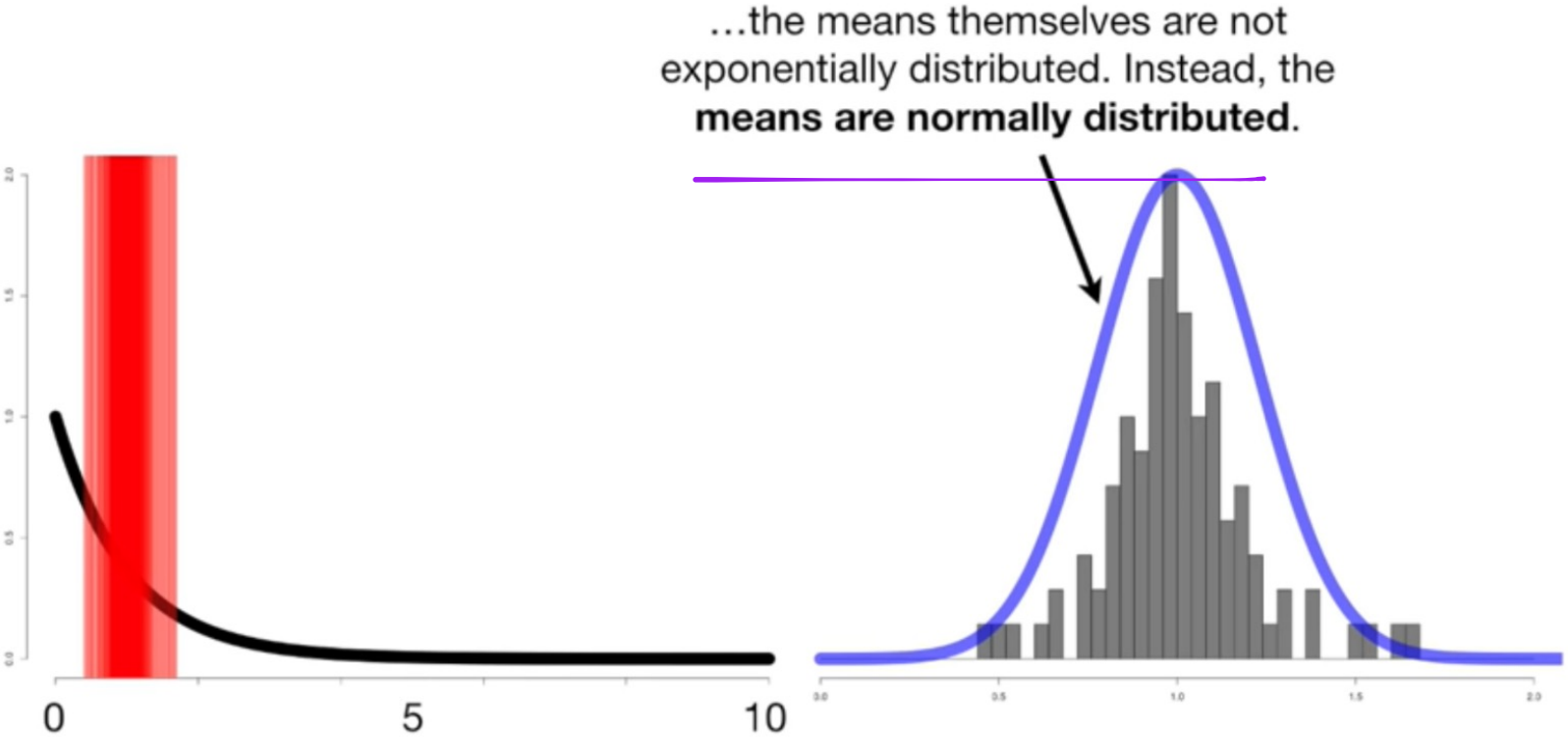
Central Limit Theorem



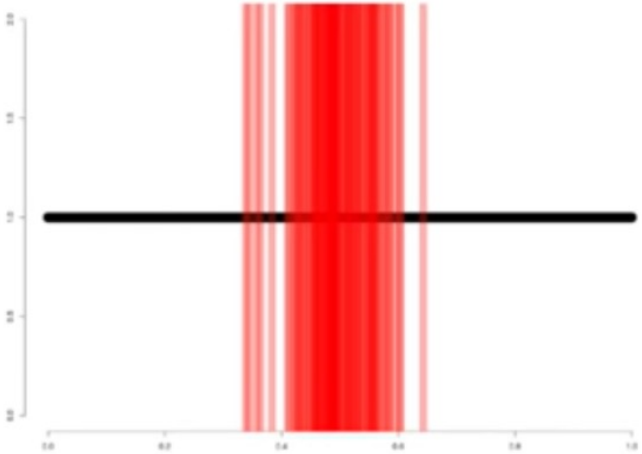
Central Limit Theorem



Central Limit Theorem

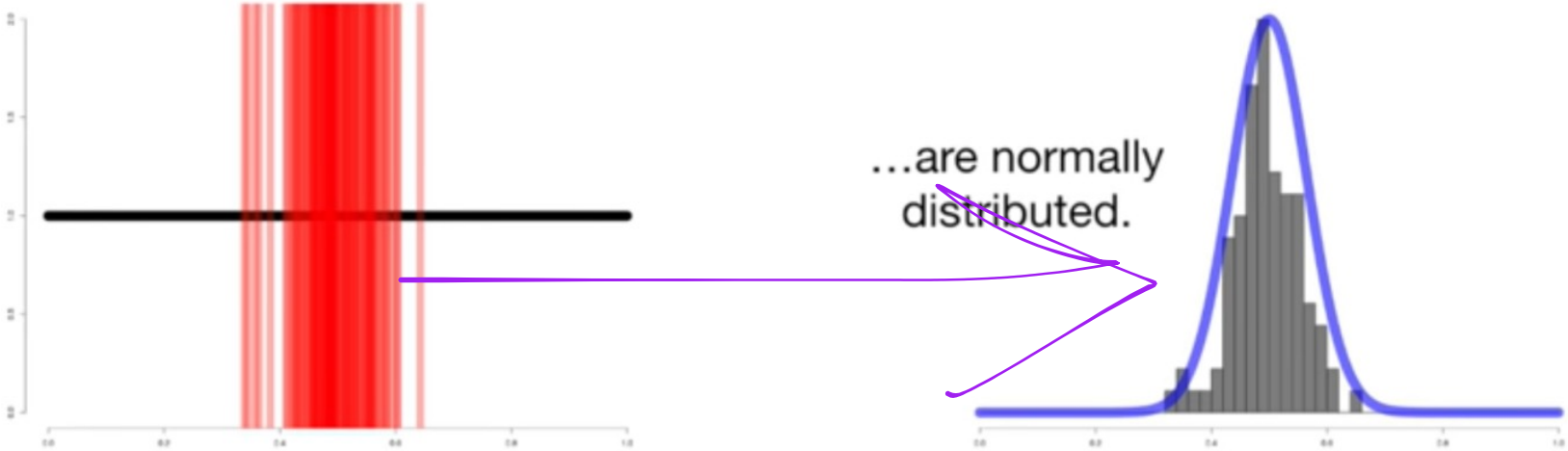


Central Limit Theorem

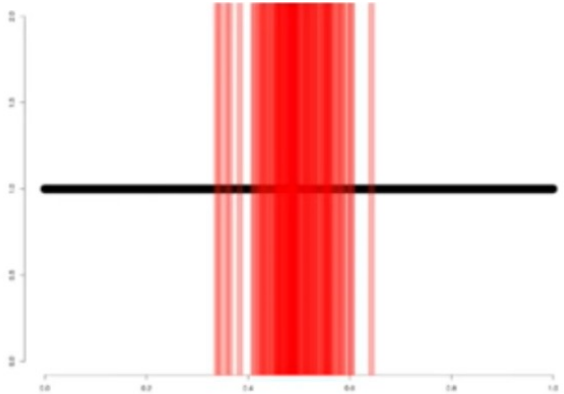


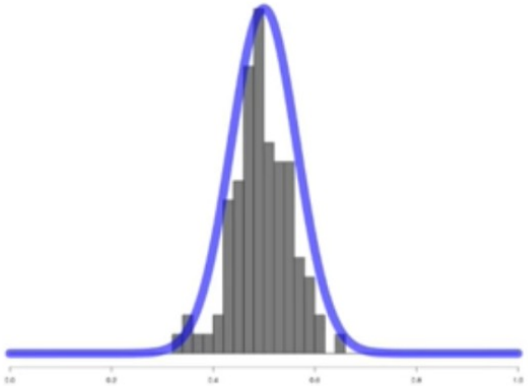
So far we have seen that means calculated from samples taken from a uniform distribution…

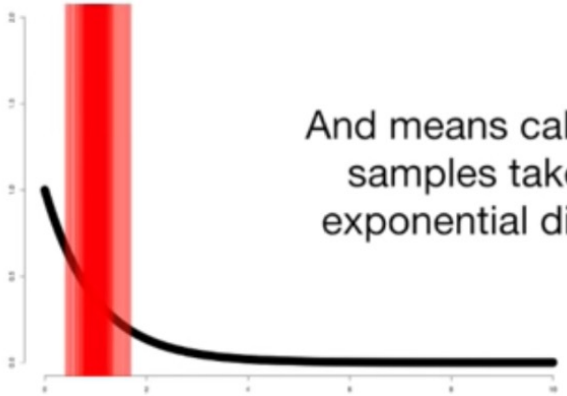
Central Limit Theorem



Central Limit Theorem

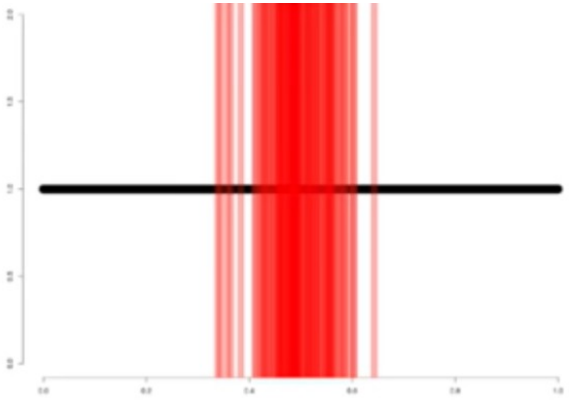


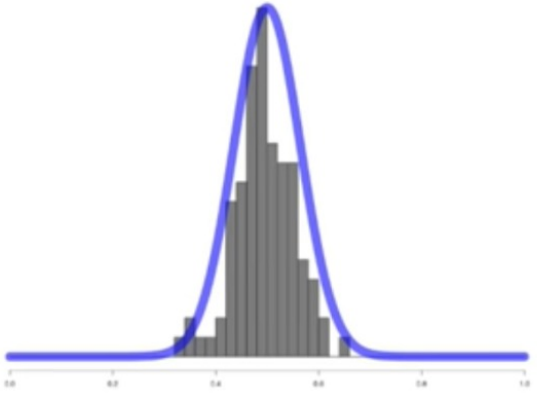


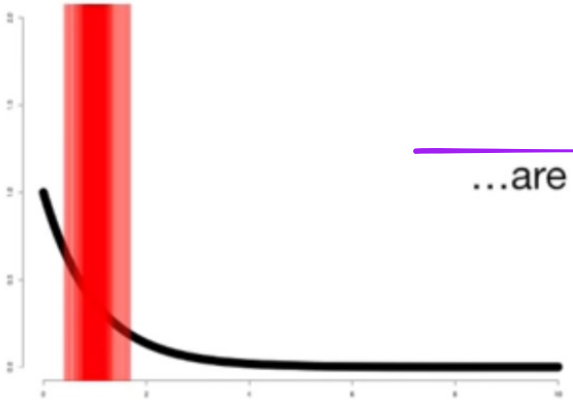


And means calculated from samples taken from an exponential distribution…

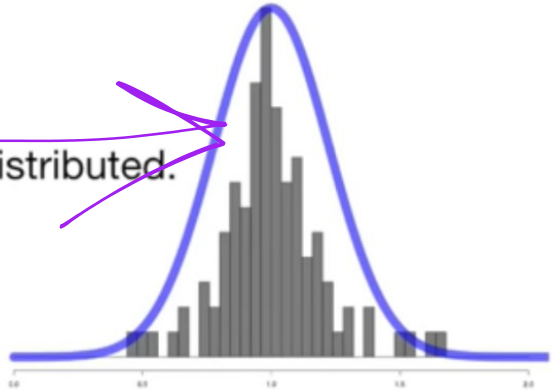
Central Limit Theorem





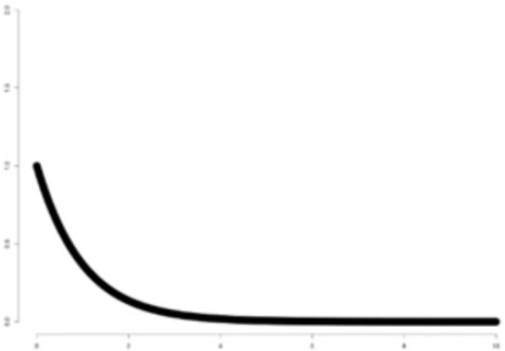


…are also normally distributed.

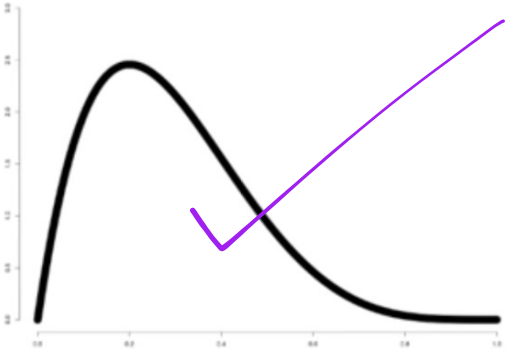


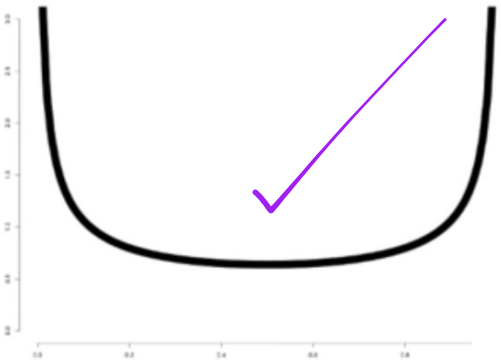
Central Limit Theorem



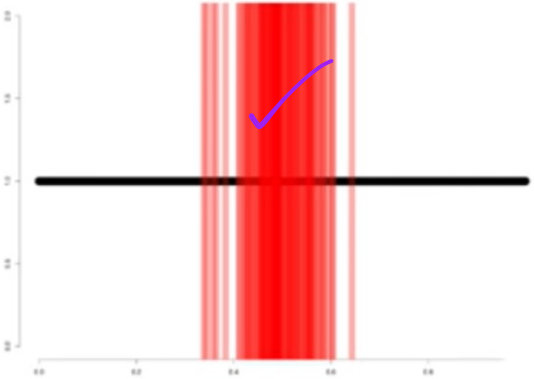


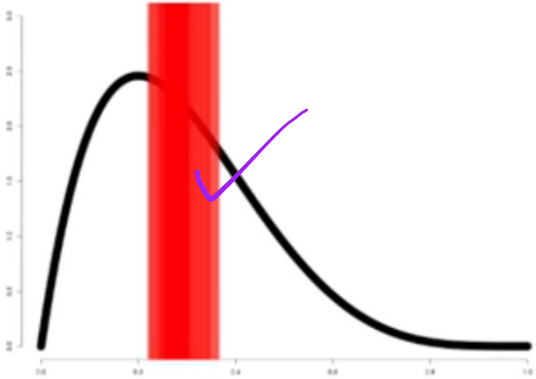
Well, it turns out that it doesn't matter what distribution you start with…



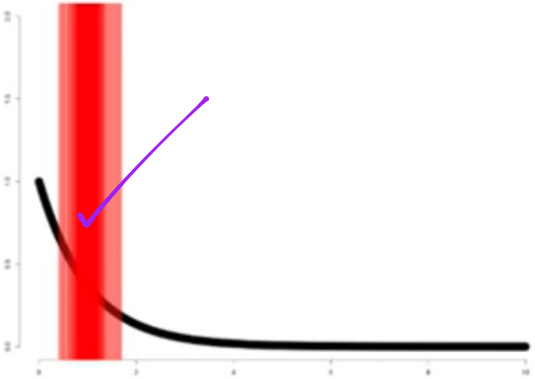


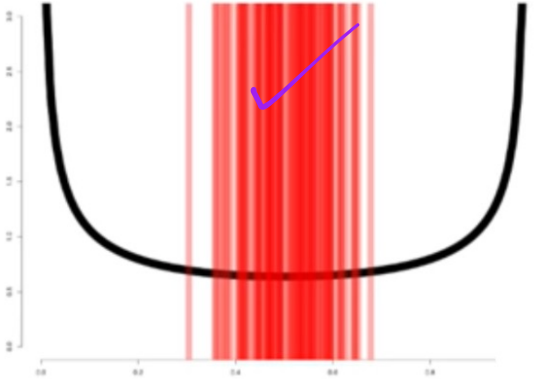
Central Limit Theorem



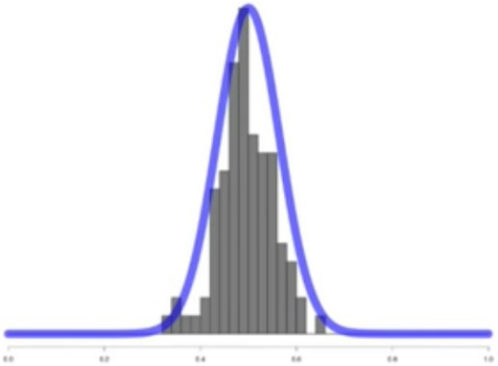


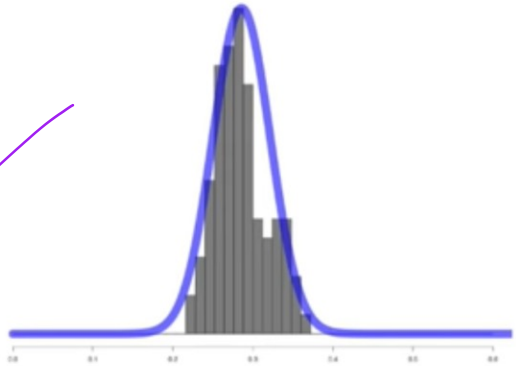
…if you collect samples from those distributions…



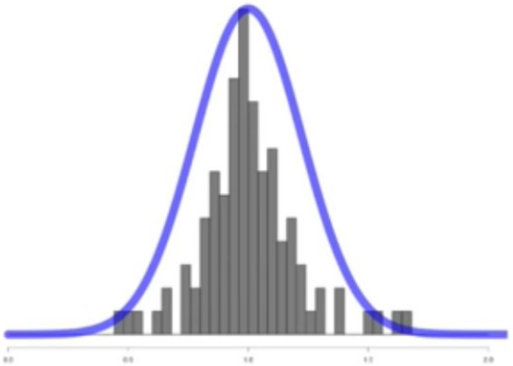


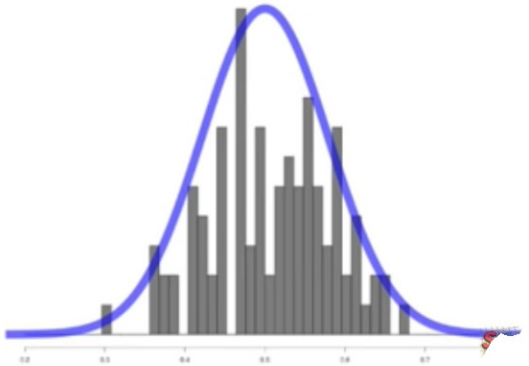
Central Limit Theorem





…then **the means will be normally distributed***.



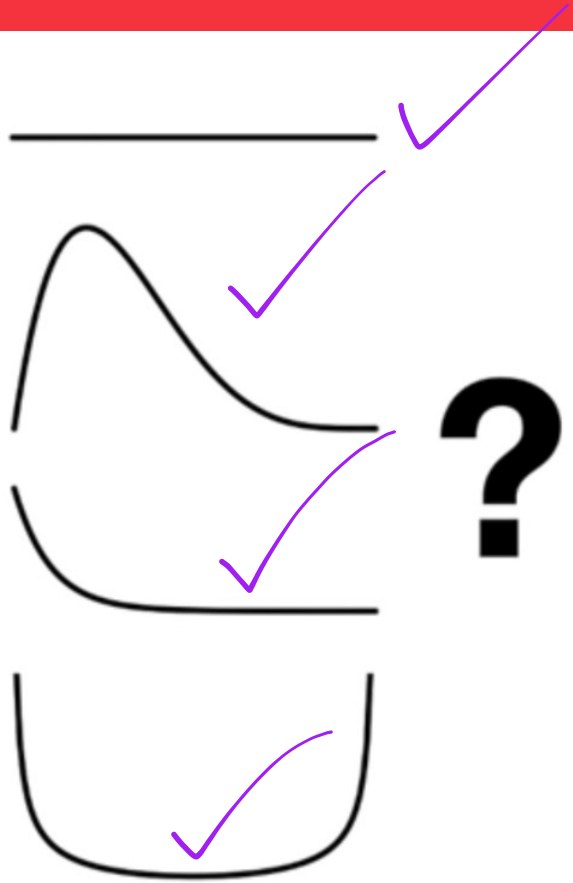


Central Limit Theorem

Cool!

But what are the practical implications of knowing that means are normally distributed?

Central Limit Theorem



?

When we do an experiment, we don't always know what distribution our data comes from.

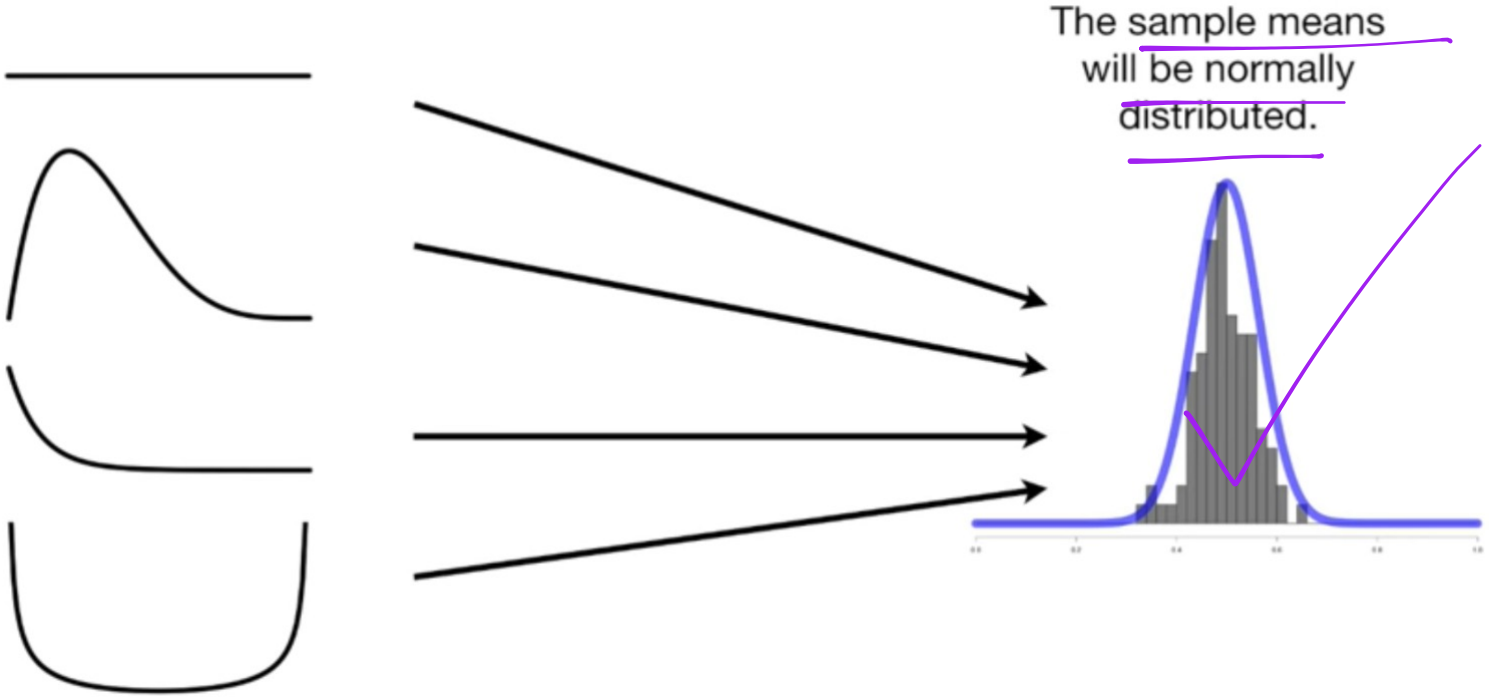
Central Limit Theorem



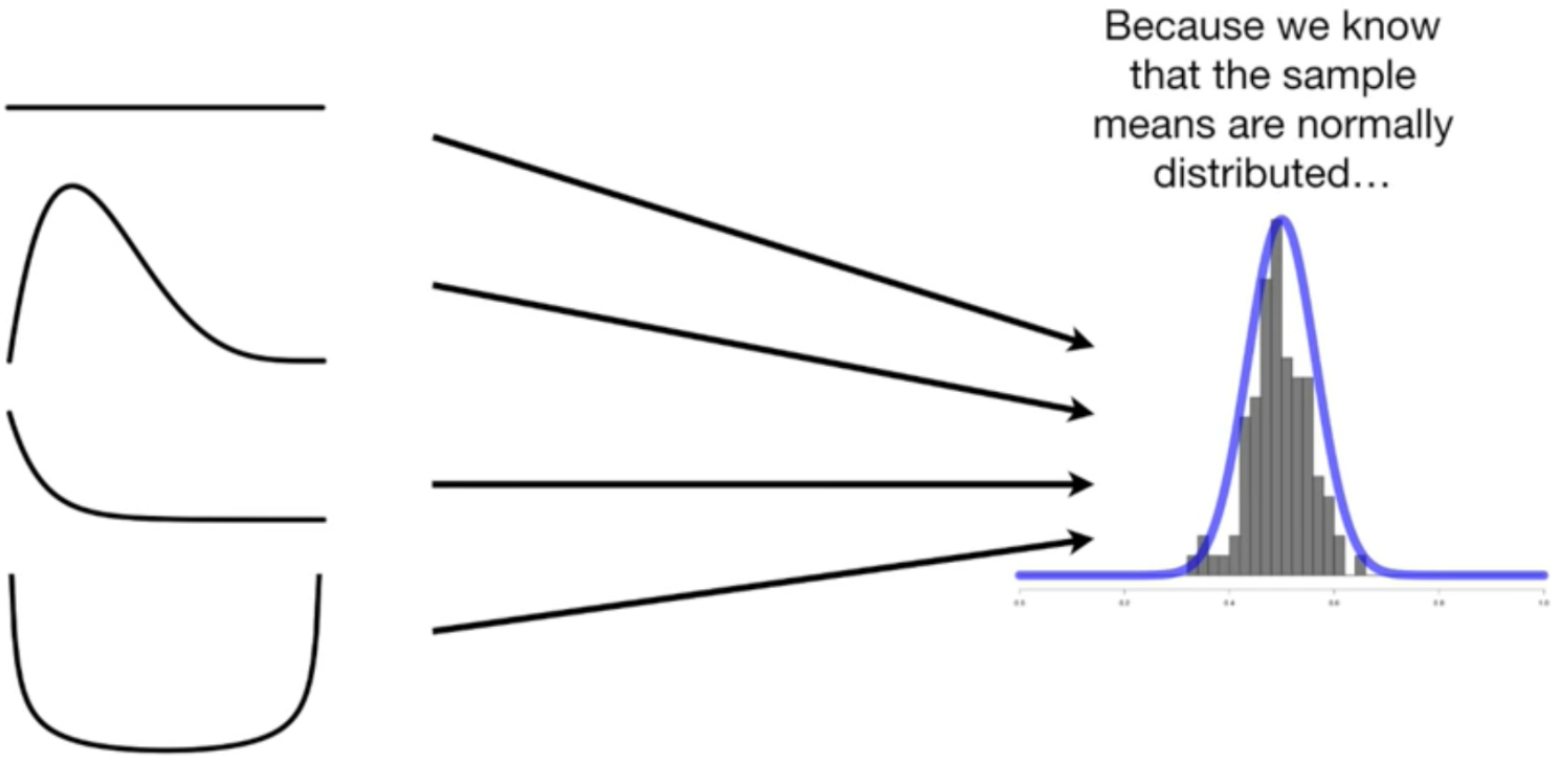
?

To this, **The Central Limit Theorem**
says, **“Who Cares???”**

Central Limit Theorem



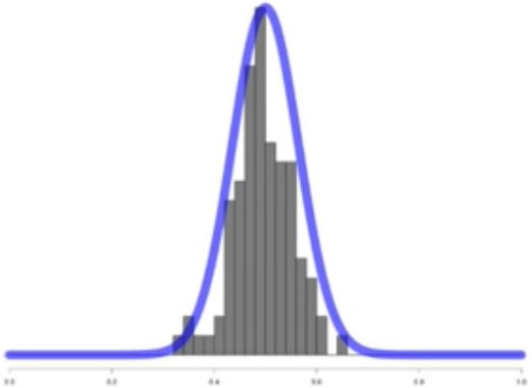
Central Limit Theorem



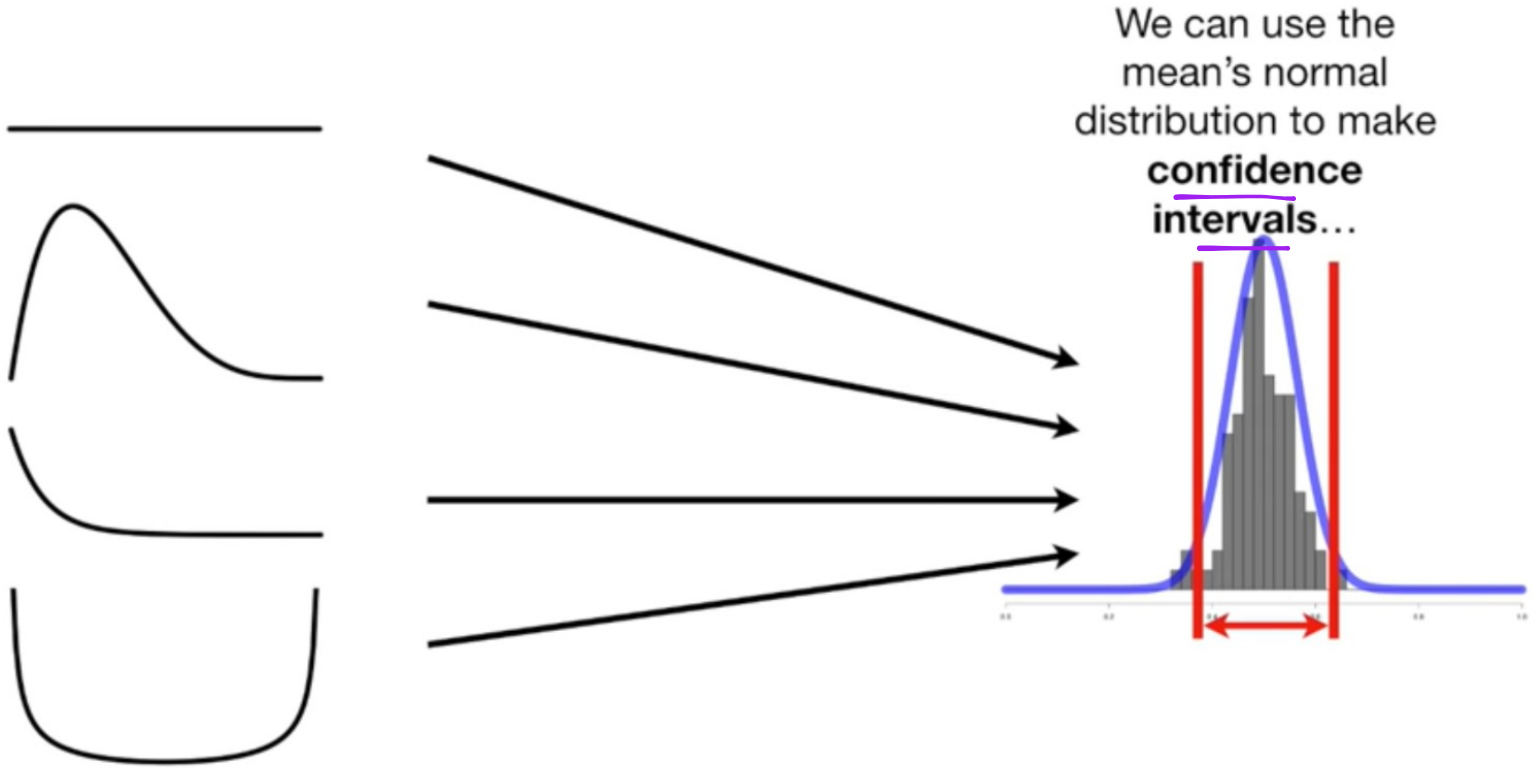
Central Limit Theorem



…we don't need to worry too much about the distribution that the samples came from.



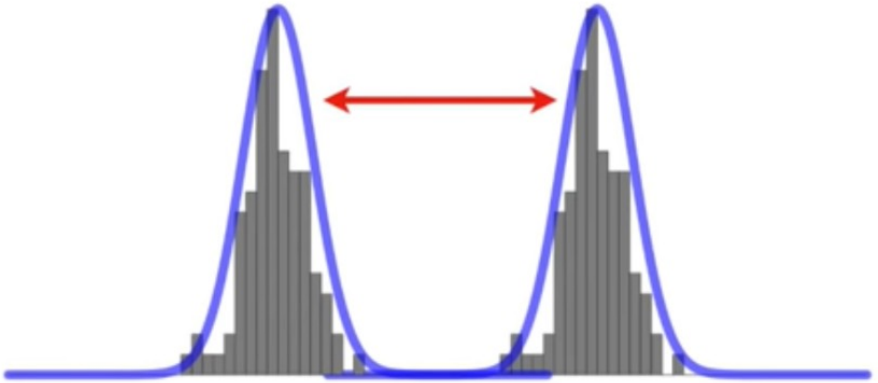
Central Limit Theorem



Central Limit Theorem



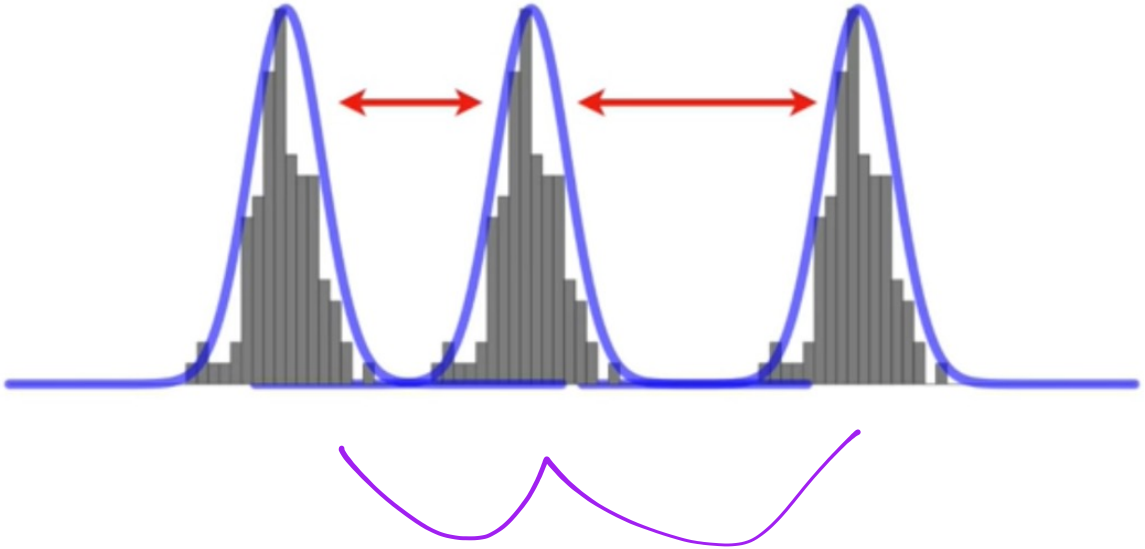
…do **t-tests**, where we ask if there is a difference between the means from two samples…



Central Limit Theorem



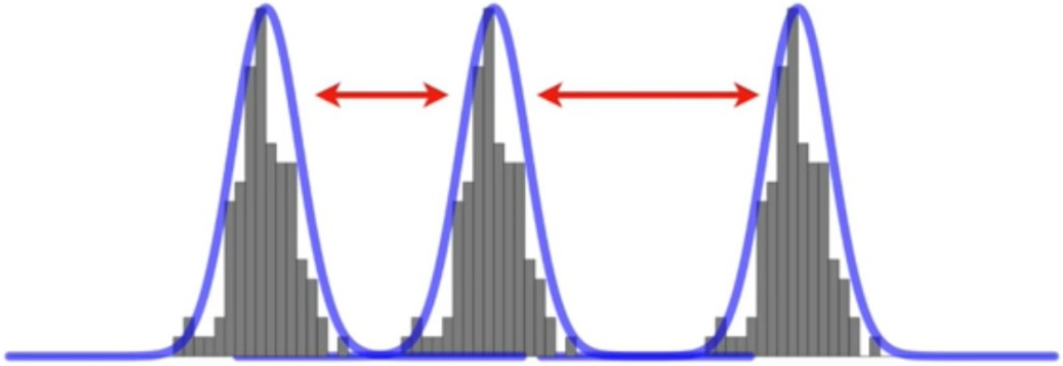
…and **ANOVA**, where we ask if there is a difference among the means from three or more samples…



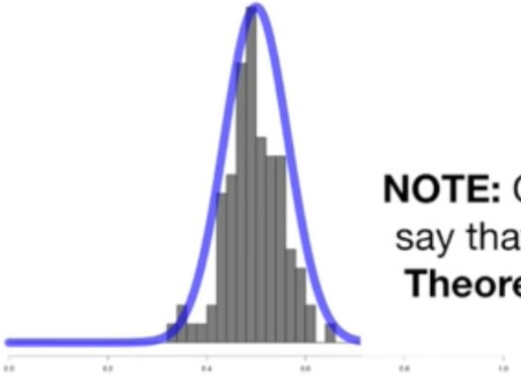
Central Limit Theorem



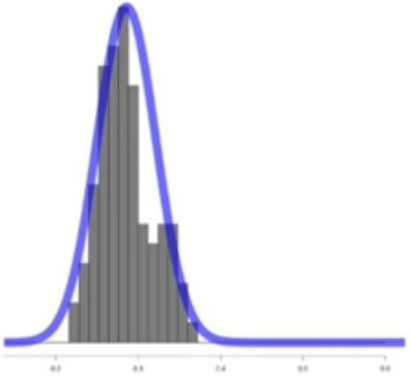
…and pretty much any statistical test
that uses the sample mean.

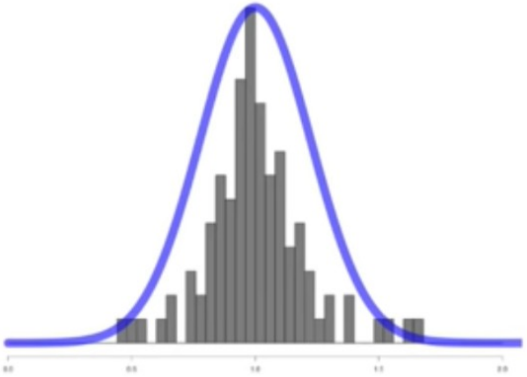


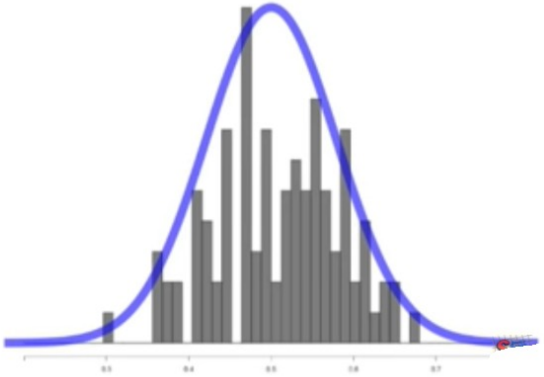
Central Limit Theorem



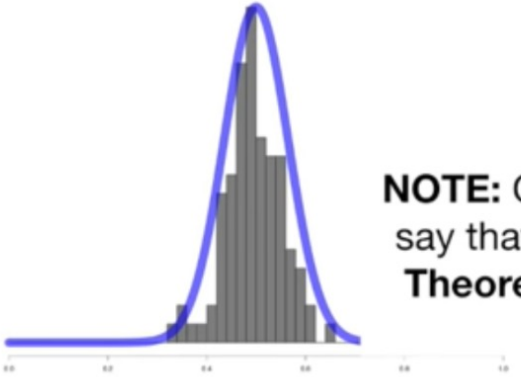
NOTE: Out there in the wild some folks say that in order for the **Central Limit Theorem** to be true, the sample size must be at least **30**.



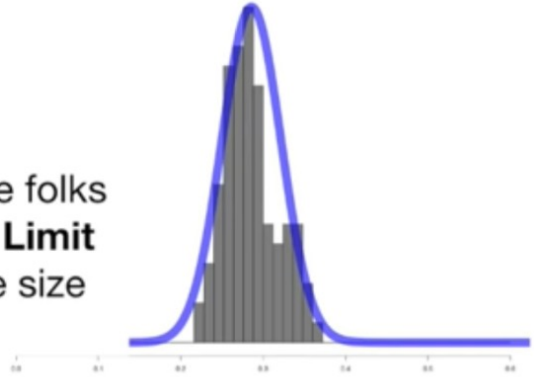




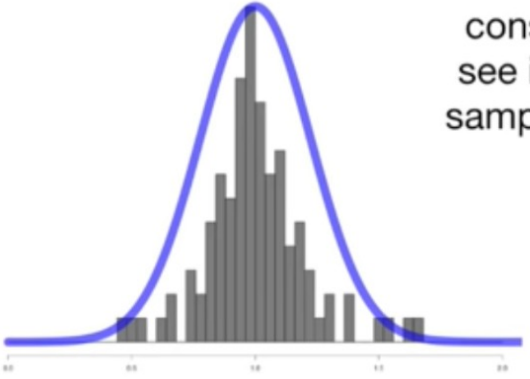
Central Limit Theorem

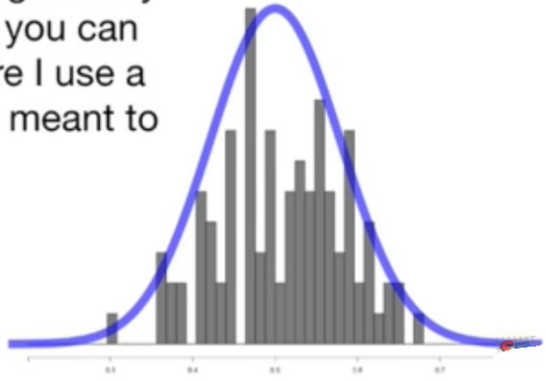


NOTE: Out there in the wild some folks say that in order for the **Central Limit Theorem** to be true, the sample size must be at least **30**.



This is just a rule of thumb and generally considered safe. However, as you can see in the examples here where I use a sample size of **20**, the rule was meant to be broken.





#LifeKoKaroLift

Thank You!

Connect me: https://www.linkedin.com/in/dr-darshan-ingle-corporate-trainer/

